The Divide-and-Conquer Estimator for High-Dimensional Linear Regression under Weak Exogeneity

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Abstract

Under weak exogeneity, ordinary least squares is inconsistent when the number of regressors is proportional to the sample size and regressors are autocorrelated, as shown by Mikusheva and Sølvsten (2025). Their consistent estimator relies on homoskedasticity. I propose the Divide-and-Conquer (DaC) estimator, which is consistent under heteroskedasticity when feedback from the error term to future regressors has a known, finite horizon. The idea is simple: partition the sample into subsamples, spaced so that strict exogeneity holds within each, estimate OLS on every subsample, and average. The resulting estimator is unbiased and consistent for linear contrasts of the slope in the proportional regime. Asymptotic normality and consistent variance estimation are established when the number of regressors grows but remains a vanishing fraction of the sample size. Simulations compare DaC, OLS, and the Mikusheva–Sølvsten estimator.

1 Introduction

In linear time series regressions, identification often relies on the assumption of weak exogeneity. Weak exogeneity requires the error term to be mean independent from the past and current regressors only and thus allows for the conditional mean of the error term to depend on future values of the regressor. Such dependence between the error term and the regressors of future periods might arise as a consequence of *feedback* from the error term to the regressor, i.e. the error term might affect the regressor in the following periods. The Ordinary Least Squares estimator has been known to have a finite-sample bias under weak exogeneity at least since Stambaugh (1999). However, in the standard asymptotic framework, this bias vanishes - OLS is consistent. Thus OLS is commonly used to estimate time series regressions even where weak exogeneity is suspected as the bias is thought to be negligible when the sample size is large enough. Mikusheva and Sølvsten (2025) show that, under weak exogeneity, OLS is inconsistent in an asymptotic framework where the number of regressors is proportional to the sample size if regressors are in addition autocorrelated. Thus, the bias of OLS may be non-negligible if the number of regressors relative to the number of observations (the *aspect ratio*) is non-negligible. Mikusheva and Sølvsten (2025) propose an estimator for

weakly exogenous regression models that is consistent under this proportional regime but relies on an assumption of homoskedasticity.
In this paper, I propose an estimator that is consistent in the proportional regime with heteroskedasticity under an exogeneity assumption that is slightly stronger than classical weak exogeneity: I allow for feedback from the error term to future regressors with a *finite horizon* of h periods only. That is, I assume the error term at time t to be mean independent with respect to regressors from the past, the present and the future - except those from periods $t+1$ through $t+h$. If the magnitude of feedback declines fast enough, this might be a reasonable approximation to the true data generating process. The idea of the estimator is simple: Under the assumption of a finite feedback horizon, if we pretend h is known, the sample can be partitioned into $h+1$ subsamples within each of which strict exogeneity holds. This is done by skipping h periods between observations. If each subsample still contains at least as many observations as variables, we can obtain an OLS estimate for each subsample and average these across subsamples. I call the resulting average the Divide-and-Conquer estimator (DaC). Due to strict exogeneity within subsamples, the Divide-and-Conquer estimator is unbiased. Adopting assumptions that largely mirror those of Mikusheva and Sølvsten (2025), I show it to be consistent for individual contrasts of the slope parameter under the proportional regime. To obtain standard inference results, however, it is necessary to either assume normal errors or to relinquish the proportional regime. Without assuming normal errors, asymptotic normality of individual contrasts is possible in an asymptotic framework where the number of regressors can grow with the sample size but only as long as it is a vanishing fraction of the sample size. I also provide simulation results that compare OLS, the Divide-and-Conquer estimator and the estimator of Mikusheva and Sølvsten (2025) (which I will henceforth refer to as the M&S estimator). While the M&S estimator generally performs well even under heteroskedasticity, it does exhibit a non-negligible bias when heteroskedasticity is combined with heavy-tailed regressors.

This paper is most closely related to Mikusheva and Sølvsten (2025). The reader is referred to that paper for an intuitive explanation of the failure of OLS under the proportional regime. While the M&S estimator (just like the DaC estimator) requires the econometrician to specify a feedback horizon, this horizon can in principle be chosen to be large, whereas the maximal horizon supported by the DaC is limited by the aspect ratio as every subsample must contain more observations then there are regressors. On the other hand, Mikusheva and Sølvsten (2025) derive inference results only for the version of their estimator for a feedback horizon equal to one period, while the current paper provides inference results for any feedback horizon which satisfies the requirement that the subsamples contain more observations than regressors.

The problem of bias in OLS due to weak exogeneity was analysied in Stambaugh (1999), who provides the distribution of OLS under concrete parametric assumptions. Sawa (1978) works out the exact moments of OLS for first-order autoregressive processes (which are subject to weak exogeneity by construction).

The other theme of this paper is high-dimensional regression. Results on the asymptotics of OLS in the proportional regime exist only under highly restrictive assumptions on the distribution of the regressors. Lei, Bickel, and El Karoui (2018) provide conditions on the asymptotic behaviour of a (non-random) design matrix for asymptotic normality of an individual regressor or a linear combination of a fixed number of regressors in the propor-

tional regime. They then provide DGPs for the regressor under which these conditions are met with high probability. These DGPs are very restrictive and, in particular, don't allow for time-series dependence. Relevant results from the statistical literature that could be applied to time series regressions are scarce. In the current paper, I therefore take the approach of Mikusheva and Sølvsten (2025) to treat the regressors as non-random and impose the necessary assumptions on the the eigenvalues of the Gram matrix directly.Dobriban and Sheng (2021) show that in the cross-sectional framework with stochastic but iid regressors, iid error terms (and strict exogeneinity) that the divide-and-conquer estimator is less efficient that the standard OLS estimator when the dimension is proportional to the sample size. (While there is no asymptotic loss of efficiency in the standard asymptotic framework with a constant number of regressors.)

Lastly, the current paper touches on the problem of estimating the variance of the OLS estimator in the proportional regime in the presence of heteroskedasticity. Cattaneo, Jansson, and Newey (2018) show that the usual Eicker-Huber-White standard errors are not consistent under the proportional regime and neither are its modified versions HC1-4. An estimator was proposed by Cattaneo, Jansson, and Newey (2018), which however requires the aspect ratio to be smaller than 1/2. Here, I show that the estimator of Jochmans (2022) works within the framework of the current paper.

General notation. I use C and c as generic constants throughout, that may take different values from application to application. For positive $a \in \mathbb{R}$, I use the notation $\lfloor a \rfloor$ to denote the largest integer weakly smaller than a. For integers $a, b \in \mathbb{N}, a \bmod b$ denotes the remainder of integer division of a by b, i.e. $a \bmod b = a - \lfloor a/b \rfloor \times b$. Let $u(t) = (t-1) \bmod (h+1)+1$ be the subsample period t belongs to and let $r(v,t) = v+(t-1)(h+1)$ be the period the t-th observation from subsample v corresponds to.

Matrix notation. In the following, for a square matrix A, $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ denote its largest and smallest eigenvalue, respectively, and $\operatorname{tr}(A)$ denotes its trace (the sum of its diagonal elements). $\|\cdot\|$ denotes the Euclidean norm when applied to vectors. Let $\|\cdot\|_2$ denote the spectral norm, i.e. for a $p \times q$ matrix B, $\|B\|_2 := \sup_{v \in R^q, \|v\| \leq 1} \|Bv\| = \sqrt{\lambda_{\max}(B'B)}$. As a consequence, when B is symmetric and positive semi-definite, $\|B\|_2 = \lambda_{\max}(B)$. (More generally, when B is symmetric, $\|B\|_2 = \max\{\lambda_{\max}(B), -\lambda_{\min}(B)\}$.) Let $\|\cdot\|_F$ denote the Frobenius norm, i.e. for a $p \times q$ matrix B, $\|B\|_F := \sum_{i,j}(B_{ij})^2 = \sqrt{\operatorname{tr}(B'B)}$. Many useful inequalities linking these norms and eigevnlues can be found in appendix B.

2 The Divide-and-Conquer estimator

We observe a sample of size T from the model

$$y_t = x_t'\beta + \varepsilon_t$$

where the dimension of x_t is denoted by K. We are interested in doing inference on the slope parameter $\beta \in \mathbb{R}^K$, or more precisely on some linear contrast of the slope parameter$r'\beta$ (for some $r \in \mathbb{R}^K$). Strict exogeneity may not hold but suppose that the feedback has a known, finite horizon of h periods, as formalized by the following assumption:

Assumption *Finite Feedback Horizon.* *There is a* $h \in \mathbb{N}$ *such that*

$$\mathbb{E}\left[\varepsilon_t | ..., x_{t-1}, x_t, x_{t+h+1}, x_{t+h+2}, ...\right] = 0$$

for all $t \in \mathbb{Z}$.

Then, strict exogeneity holds within all subsamples where all observations are at least $h+1$ periods apart from each other. We can partition the sample into $h+1$ subsamples sampled at regular intervals. Thus we obtain the subsamples $D_1 = \{1, h+2, 2h+3, \dots\}, D_2 = \{2, h+3, 2h+4, \dots\}, \dots, D_{h+1} = \{h+1, 2h+2, \dots\}$. For ease of notation we suppose that T is a multiple of $h+1$ so that each subsample has the same size. Let $\bar{T} = T/(h+1)$ denote this subsample size. We denote the submatrix of observations of the regressors from subsample $v \in \{1, \dots, h+1\}$:

$$\mathbf{X}_v = \begin{pmatrix} x_v' \\ x_{v+h+1}' \\ x_{v+h+2}' \\ \vdots \end{pmatrix}$$

Likewise, the subvector of outcome values for periods belonging to subsample $v \in \{1, \dots, h+1\}$ is $Y_v = (y_v, y_{v+h+1}, y_{v+2h+2}, \dots)'$.

Suppose that the subsample size exceeds the dimension, i.e. $\bar{T} = T/(h+1) > K$ so that we can obtain an OLS estimator for each subsample:

$$\hat{\beta}_v = \left(\mathbf{X}_v'\mathbf{X}_v\right)^{-1}\mathbf{X}_v'Y_v$$

Each of these is an unbiased estimator of β so their average

$$\hat{\beta} = \frac{1}{h+1}\sum_{v=1}^{h+1}\hat{\beta}_v$$

is unbiased as well. I call $\hat{\beta}$ the Divide-and-Conquer estimator.

3 Asymptotic Theory

To obtain asymptotic results about the DaC that can be relied upon in high-dimensional settings, I adopt an asymptotic framework in which the number of regressors grows with the sample size. More specifically, I consider a sequence of data generating processes - one for each sample size T - a generally different set of regressors for each sample size and potentially even a different outcome variable. This sequence of DGPs is controlled by imposing bounds on the eigenvalues of the Gram matrix. Although K, β, now depend on T, I suppress the subscript T to keep notation simple. We also limit attention to one linear contrast $r'\beta$ for some sequence $r \in \mathbb{R}^K$.

To deal with a framework that combines increasing dimension and regressors that are not fully exogenous, I follow the framework of Mikusheva and Sølvsten (2025). In particular, the regressors can be decomposed into a strictly exogenous part and feedback from past periods. Like Mikusheva and Sølvsten (2025), I treat the strictly exogenous part of the regressors as non-random.

Assumption *Feedback Decomposition*. *The observed regressors x_t admit the decomposition*

$$x_t = \tilde{x}_t + \sum_{\ell=1}^{h} \alpha_\ell \varepsilon_{t-\ell},$$

where the K-vectors $\tilde{x}_1, \ldots, \tilde{x}_T$ are non-random, the K-vectors $\alpha_1, \ldots, \alpha_h$ are non-random and $\mathbb{E}[\varepsilon_t] = 0$. h is fixed (does not depend on T).

The vectors $\alpha_1, \ldots, \alpha_h$ provide the feedback strength for each regressor at each feedback horizon. They too depend on the sample size. We write $\tilde{\mathbf{X}}_v = (\tilde{x}_v, \tilde{x}_{v+h+1}, \ldots)'$ for the matrix assembling the regressors of each subsample v. The next assumption provides bounds on the sequence of data generating processes. The eigenvalue bounds on $\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}$ (the subsample Gram matrices) ensure that variances of the regressors do not grow without bound and do not vanish either.

Assumption *Bounded Eigenvalues*. *For all T :*

$$\infty > C > \lambda_{\max}\left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}\right) \geq \lambda_{\min}\left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}\right) > c > 0$$

for all subsamples. $\|r\|, \|\alpha_1\|, \ldots, \|\alpha_h\| < C$ for all T.

To assess the plausibility of this assumption when $\tilde{\mathbf{X}}_v$ is random, imagine for a moment that the individual entries of each $\tilde{\mathbf{X}}_v$ are i.i.d. standardized random variables with bounded fourth moments and that $K/\bar{T} \to c \in (0,1)$. In this case, the empirical spectral distribution of $\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}$ converges to a Marcenko-Pastur law (Marčenko and Pastur (1967)) , whose support is $[(1-\sqrt{c})^2, (1+\sqrt{c})^2]$. Furthermore, with probability one, no eigenvalue of $\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}$ lies outside this limiting support for all large $\bar{T}$ (Bai and Silverstein (1998)). Only limited generalizations of results like these to data that not i.i.d across variables and observations exist. Therefore, I sidestep the random matrix aspect by treating the exogenous part of the regressors as fixed.

Regarding the error term, I allow for heteroskedasticity within bounds and require uniformly bounded fourth moments. As the $\tilde{x}_t$ are treated as non-random, the relevant notion of heteroskedasticity is unconditional heteroskedasticity. I assume error distributions to be independent and to differ only by their scale.

Assumption *Regular Errors*. *$0 < c < \sigma_t^2 < C < \infty$ for all t. The normalized errors $\{\varepsilon_t/\sigma_t\}_t$ are iid and $\mathbb{E}[\varepsilon_t^4] \leq C^2$ for all t.*

The following assumption allows the aspect ratio to be proportional to sample size subject to the constraint that there be more than K observations in each subsample.

Assumption *Moderate-Dimensional Regime*. $\limsup \frac{K}{T} < \frac{1}{h+1}$.

Under these conditions, the DaC is consistent for the linear contrast $r'\beta$:

Theorem 1. *Under assumptions* Feedback Decomposition*,* Bounded Eigenvalues*,* Regular Errors *and* Moderate-Dimensional Regime*,*

$$r'(\hat{\beta} - \beta) \xrightarrow{p} 0.$$

We next turn to the question of inference and in particular asymptotic normality of the DaC estimator. When assuming the decomposition of X stipulated by assumption *Feedback Decomposition* into a non-random, strictly exgoneous part $\tilde{\mathbf{X}}$ and feedback, the DaC estimator can asymptotically be written as a combination of a linear form in errors $\omega'\varepsilon_t$ (where $\omega = (\omega_1, \ldots, \omega_T)$ is some deterministic vector of weights) and a quadratic form in the error term $\varepsilon_t' W \varepsilon_t$ (where W is some deterministic matrix of weights). Asymptotic normality of the normalised linear form hinges on the condition that no individual weight dominate the overall variance of the linear form, i.e. $\max_t \omega_t = o(\|\omega\|)$. Towards formulating the necessary assumptions, we define the following::

Definition 1. Let $\mathbf{A} := (r, \alpha_1, \ldots, \alpha_h)' \in \mathbb{R}^{(h+1)\times K}$ and define the cross-leverage vectors

$$f_t := \mathbf{A}(\tilde{\mathbf{X}}_{u(t)}'\tilde{\mathbf{X}}_{u(t)}/\bar{T})^{-1}\tilde{x}_t \ \in \ \mathbb{R}^{h+1}, \qquad t = 1, \ldots, T.$$

Within our framework, the condition that no individual weight dominate the linear form amounts to the following assumption on f_t:[1]

Assumption *No Influential Observations*. A *has full row rank uniformly in T, i.e.* $\lambda_{\min}(\mathbf{A}\mathbf{A}') \geq c > 0$, *and*

$$\max_{1\leq t\leq T} \|f_t\| = o(T^{1/2}).$$

When $h = 0$, so that $\tilde{\mathbf{X}}_v = \mathbf{X}_v$, Assumption *No Influential Observations* reduces to $\max_t r'(\mathbf{X}_{u(t)}'\mathbf{X}_{u(t)})^{-1}x_t = o(T^{-1/2})$. This condition corresponds to the necessary and sufficient condition from Lemma 2.1 in Huber (1973) and ensures that the influence of the most influential observation vanishes. When $h \geq 1$, the same must hold jointly for the r-direction and all h feedback directions. While theoretical results relevant to showing convergence in probability of $T^{-1/2}\max_t \|f_t\|$ in the proportional regime under general assumptions (in particular, under time series assumptions) when treating $\tilde{x}_t$ as random are not available, simulation evidence in appendix C suggests that this holds at least for the particular DGP simulated (the exogenous part of the regressors are generated as a stationary VAR.) A sufficient condition for assumption *No Influential Observations* to hold is that the maximum leverage $\max_t \tilde{x}_t'(\tilde{\mathbf{X}}_{u(t)}'\tilde{\mathbf{X}}_{u(t)})^{-1}\tilde{x}_t$ vanishes (which is only possible when $K/\bar{T} \to 0$, however).

The quadratic part of $r'(\hat{\beta} - \beta)$, on the other hand, is asymptotically normal without additional assumptions. A key property of the DaC estimator is that the diagonal elements of this quadratic form are each individually 0 (which is what makes it unbiased). This contrasts with the M&S estimator, where the diagonal elements of the associated quadratic form are (asymptotically) 0 only on average (which makes it (asymptotically) unbiased under homoskedastitcy). Having each individual diagonal element equal to zero facilitates inference. Thus, we obtain the following result:

[1] A weaker but more technical alternative to assumption *No Influential Observations* is the following: Let $\bar{A}$ be a full row rank matrix which has the same row space as A. As an alternative to assumption *No Influential Observations*, assume

$$\max_t \tilde{x}_t'(\tilde{X}_{u(t)}'\tilde{X}_{u(t)}/\bar{T})^{-1}\bar{A}'(\bar{A}(\tilde{X}_{u(t)}'\tilde{X}_{u(t)}/\bar{T})^{-1}\bar{A}')^{-1}\bar{A}(\tilde{X}_{u(t)}'\tilde{X}_{u(t)}/\bar{T})^{-1}\tilde{x}_t = o(\bar{T}).$$

The advantage of this assumption is that it does not require $\lambda_{\min}(AA') > c$.

Theorem 2. *Under assumptions* Feedback Decomposition*,* Bounded Eigenvalues*,* Regular Errors*,* Moderate-Dimensional Regime*, and* No Influential Observations*, for some scalar sequence* Σ *defined in the proof of this theorem,*

$$\frac{r'(\hat{\beta}-\beta)}{\sqrt{\Sigma/(h+1)^2}} \xrightarrow{d} N(0,1).$$

By contrast, inference results for the M&S estimator in the proportional regime are only available under the additional assumption of normally distributed errors.

We now turn to estimating the asymptotic variance $\Sigma/(h+1)$. When K is proportional to T, the Eicker-Huber-White (HC0) variance estimator is no longer consistent and the same holds for all other members of the HCk class (Cattaneo, Jansson, and Newey (2018)), even when focussing on just one or a constant number of linear combinations of the parameter. The HC0-estimator simply estimates each σ_t^2 by $\hat{\varepsilon}_t^2$. If σ_t^2 could instead be estimated by ε_t^2, the resulting estimator would be consistent even in the proportional regime (under a somewhat strengthened condition on the most influential observation, see Assumption *Bounded Cross-Leverage* below). However each $\hat{\varepsilon}_t^2$ depends not only on ε_t but also on the other ε_j. When $K/\bar{T}$ does not vanish, the influence of the most influential observation's ε_j on each other observation's $\hat{\varepsilon}_t$ does not vanish because maximum leverage $\max_t \tilde{x}_t(\tilde{\mathbf{X}}'_{u(t)}\tilde{\mathbf{X}}_{u(t)})^{-1}\tilde{x}_t$ is always larger than $K/\bar{T}$. For the HC0 estimator, this entails an asymptotic downward bias (which can also be regarded as a result of overfitting), whereas the $HC3$ estimator (the jackknife estimator of MacKinnon and White (1985)) is asymptotically upward biased (i.e. conservative). Several alternative heteroskedasticity-robust variance estimators have been proposed for the proportional regime. Cattaneo, Jansson, and Newey (2018) propose a variance estimator (resembling the one of Bera, Suprayitno, and Premaratne (2002)) which replaces the $\hat{\varepsilon}_t^2$ by linear combinations of all the $\hat{\varepsilon}_j^2$ in such a way that the effects of the $\hat{\varepsilon}_j^2$ for $j \neq t$ cancel when taking the conditional expectation. That is, we consider an estimator of the form

$$\hat{\Sigma} = \sum_{v=1}^{h+1} r'(\mathbf{X}'_v\mathbf{X}_v)^{-1} \sum_{t=1}^{\bar{T}} x_{r(v,t)}x'_{r(v,t)} \left(\sum_{j=1}^{\bar{T}} \kappa_{v,tj} \right) \hat{\varepsilon}^2_{r(v,j)}(\mathbf{X}'_v\mathbf{X}_v)^{-1}r$$

where the $\kappa_{v,j,t}$ are weights that depend on the regressors of each subsample and are chosen such that $\mathbb{E}\left[\sum_{j=1}^{\bar{T}} \kappa_{v,t,j}\hat{\varepsilon}^2_{r(v,j)}|\mathbf{X}_v\right] = \sigma^2_{r(v,t)}$. Let $\boldsymbol{H}_v = \mathbf{I}_{\bar{T}} - \mathbf{X}_v(\mathbf{X}'_v\mathbf{X}_v)^{-1}\mathbf{X}'_v$. Then, using

that $\hat{\varepsilon}_{r(v,t)} = \sum_{j=1}^{\bar{T}} \mathbf{H}_{v,tj}\varepsilon_{r(v,j)}$, we have

$$\begin{aligned}
\mathbb{E}\left[\sum_{j=1}^{\bar{T}} \kappa_{v,tj}\hat{\varepsilon}^2_{r(v,j)}|\mathbf{X}_v\right] &= \sum_{j=1}^{\bar{T}} \kappa_{v,tj}\mathbb{E}\left[\hat{\varepsilon}^2_{r(v,j)}|\mathbf{X}_v\right] \\
&= \sum_{j=1}^{\bar{T}} \kappa_{v,tj}\mathbb{E}\left[\left(\sum_{l=1}^{\bar{T}} \mathbf{H}_{v,jl}\varepsilon_{r(v,l)}\right)^2 |\mathbf{X}_v\right] \\
&= \sum_{j=1}^{\bar{T}} \kappa_{v,tj}\sum_{l=1}^{\bar{T}} \mathbf{H}^2_{v,jl}\sigma^2_{r(v,l)} \\
&= \sum_{l=1}^{\bar{T}} \left(\sum_{j=1}^{\bar{T}} \kappa_{v,t,j}\mathbf{H}^2_{v,jl}\right)\sigma^2_{r(v,l)}
\end{aligned}$$

Hence choosing $\kappa_v = (\mathbf{H}_v \odot \mathbf{H}_v)^{-1}$ yields $\mathbb{E}\left[\sum_{j=1}^{\bar{T}} \kappa_{v,t,j}\hat{\varepsilon}^2_{r(v,j)}|\mathbf{X}_v\right] = \sigma^2_{r(v,t)}$, whenever $\mathbf{H}_v \odot \mathbf{H}_v$ is invertible.[2] While this ensures unbiasedness, consistency requires a bound on the norm of κ_v. To bound κ_v, a sufficient condition is

Assumption *Sub-One-Half Leverage*. *There is a $c_\kappa > 0$ such that, for all T,*

$$\max_{1\le t\le T} \tilde{x}_t'(\tilde{\mathbf{X}}'_{u(t)}\tilde{\mathbf{X}}_{u(t)})^{-1}\tilde{x}_t \;\le\; \frac{1}{2} - c_\kappa.$$

This condition also ensures that $\mathbf{H}_v \odot \mathbf{H}_v$ is invertible with probablity approaching 1. Finally,we also need to strengthen the assumption on influential observations somewhat:

Assumption *Bounded Cross-Leverage*.

$$\frac{1}{\bar{T}}\sum_{t=1}^{\bar{T}} \|f_{r(v,t)}\|^4 \le C$$

for all v.

(Note that Assumption *Bounded Cross-Leverage* implies the conditoin $\max_{1\le t\le T}\|f_t\| = o(T^{1/2})$ Assumption *No Influential Observations*.) We have the following result:

Theorem 3. *Under Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, Bounded Cross-Leverage *and* Sub-One-Half Leverage, *$\mathbf{H}_v \odot \mathbf{H}_v$ is invertible with probability approaching* 1 *and when choosing $\kappa_v = (\mathbf{H}_v \odot \mathbf{H}_v)^{-1}$, $\frac{\hat{\Sigma}}{\Sigma} \xrightarrow{p} 1$.*

[2]Note that this choice of κ_v differs slightly from the one proposed in Cattaneo, Jansson, and Newey (2018). The difference is negligible.

As $\max_{1\leq t\leq T}\ \tilde{x}_t'(\tilde{\mathbf{X}}_{u(t)}'\tilde{\mathbf{X}}_{u(t)})^{-1}\tilde{x}_t \geq K/\bar{T}$, an important implication of Assumption *Sub-One-Half Leverage* is that $K/\bar{T} < 1/2$, which may be too restrictive in many cases.
An alternative estimator was proposed by Jochmans (2022). It takes the form:[3]

$$\acute{\Sigma} = \sum_{v=1}^{h+1} r'(\mathbf{X}_v'\mathbf{X}_v)^{-1}\sum_{t=1}^{\bar{T}} x_{r(v,t)}x_{r(v,t)}'y_{r(v,t)}\frac{\hat{\varepsilon}_{r(v,t)}}{\mathbf{H}_{v,tt}}(\mathbf{X}_v'\mathbf{X}_v)^{-1}r$$

$\frac{\hat{\varepsilon}_{r(v,t)}}{\mathbf{H}_{v,tt}}$ is a leave-one-out estimator for $\varepsilon_{r(v,t)}$, as

$$\frac{\hat{\varepsilon}_{r(v,t)}}{\mathbf{H}_{v,tt}} = y_{r(v,t)} - x_{r(v,t)}'\hat{\beta}_{v,-r(v,t)},$$

where $\hat{\beta}_{v,-r(v,t)}$ is the OLS estimator from a regression in subsample v dropping period $r(v,t)$ (Kline, Saggio, and Sølvsten (2020)).

This estimator is robust to $K/\bar{T} \geq 1/2$, which however comes at the cost of two additional assumptions. The more restrictive one of these is the following:

Assumption *Bounded Fitted Values*. $\max_t(\beta' x_t)^2 = o_p(T)$

This requires the norm of β not to grow fast over successive DGPs. One way of thinking about this requirement is to consider the outcome variable to be the same one across successive DGPs while introducing more and more explanatory variables. Each of these explanatory variables then explains on average a vanishing fraction in the variation of the outcome variable.

Another requirement of this estimator is that $1/\min_t\{1 - \tilde{x}_t'(\tilde{X}_{u(t)}'\tilde{X}_{u(t)})^{-1}\tilde{x}_t\} = O_p(1)$. This condition is satisfied if we slightly strengthen our existing eigenvalue condition:

Assumption *Leave-One-Out Conditioning*. *For each subsample* $v \in \{1,\dots,h+1\}$, *define*

$$\tilde{\mathbf{X}}_{v,-t} = (\tilde{x}_{r(v,1)},\dots,\tilde{x}_{r(v,t-1)},\tilde{x}_{r(v,t-1)},\dots)'$$

($\tilde{\mathbf{X}}_v$ *with the* t*–th observation omitted). Then*

$$\lambda_{\min}\left(\frac{1}{\bar{T}}\tilde{\mathbf{X}}_{v,-t}'\tilde{\mathbf{X}}_{v,-t}\right) \geq c \quad \textit{for all } t.$$

This assumption requires the leave-one-out estimators from all the subsamples to be uniformly well-conditioned. If $\tilde{\mathbf{X}}_{v,-t}'\tilde{\mathbf{X}}_{v,-t}$ was singular while $\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v$ is full rank, this would mean that observation t always has a perfect fit regardless of the y_t so that nothing could be learned about $\varepsilon_{r(v,t)}$ or $\sigma^2_{r(v,t)}$.[4]

Theorem 4. *Under Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, Bounded Cross-Leverage, Bounded Fitted Values *and* Leave-One-Out Conditioning, $\frac{\acute{\Sigma}}{\Sigma} \xrightarrow{p} 1$.

[3]Note that this estimator differs slightly from what is proposed in Jochmans (2022) but the difference is negligible.

[4]To understand why, consider what it would mean for $\tilde{\mathbf{X}}_{v,-t}'\tilde{\mathbf{X}}_{v,-t}$ not to be invertible while $\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v$ is invertible. $\tilde{\mathbf{X}}_{v,-t}'\tilde{\mathbf{X}}_{v,-t}$ and $\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v$ are full rank if and only if $\tilde{\mathbf{X}}_{v,-t}$ and $\tilde{\mathbf{X}}_v$ are full rank, respectively. Thus, if

$$\operatorname{rank}(\tilde{\mathbf{X}}_{v,-t}'\tilde{\mathbf{X}}_{v,-t}) = \operatorname{rank}(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v) - 1 = K - 1,$$

4 Simulations

To demonstrate the bias of OLS under weak exogeneity and of the M&S estimator under heteroskedasticity and compare their performance with the Divide-and-Conquer estimator, I present simulation results for a data generating process where the feedback lasts one period ($h = 1$) and three periods ($h = 3$), respectively. While the bias of the M&S estimator can be quite small even under heteroskedasticity under some DGPs, it does exhibit a non-negligible bias when regressors are heavy-tailed. In the simulations below, the strictly exogenous parts of the regressors follow autoregressive processes whose error terms follow t-distributions with infinite fourth moments. The following tables show the bias and standard deviation of the first coefficient as estimated by OLS, M&S and DaC as approximated by Monte Carlo simulations.

The following is a description of the DGP under which the following simulations were run:

Data Generating Process

- Only the first regressor is affected by feedback.
- The strictly exogenous part of the regressors follows independent AR(1) processes for each variable with identical parameter: $\tilde{x}_{k,t} = b\tilde{x}_{k,t-1} + \eta_{k,t}$, $|b| < 1$.
- $\eta_{k,t} \sim t(\nu)\sqrt{\nu - 2/\nu}$ (where $t(\nu)$ is Student's t-distribution with ν degrees of freedom [a]) for some $4 \geq \nu > 2$. (So that $Var(\eta_{k,t}) = 1$ and $\mathbb{E}\left[\eta_{k,t}^4\right] = \infty$.)
- Error terms are then drawn as follows: Let $\{\tilde{\varepsilon}_t\}_t \overset{iid}{\sim} N(0,1)$. The error terms are

$$\varepsilon_t = \tilde{\varepsilon}_t \sigma \frac{\|\tilde{x}_t\|^p}{\sqrt{\sum_t \|\tilde{x}_t\|^{2p}/T}}$$

  for some $\sigma > 0$, $p \geq 1$ (with $\varepsilon_0 = \tilde{\varepsilon}_0\sigma$.) The factor $\|x_t\|^p$ introduces heteroskedasticity while the denominator $\sqrt{\sum_t \|\tilde{x}_t\|^{2p}/T}$ normalizes the variances to ensure that average variance is equal to σ^2.
- h–period feedback is then added to the first regressor, $x_{1t} = \tilde{x}_{1t} + \sum_{l=1}^{h} a_l \varepsilon_{t-l}$, while the remaining regressors are strictly exogenous, i.e. $x_{2,t} = \tilde{x}_{2,t}$.

[a] For $\nu \in (0,2]$, $X \sim t(\nu)$ has infinite variance and kurtosis. For $\nu \in (2,4]$, $X \sim t(\nu)$ has $Var(X) = \frac{\nu}{\nu-2}$ and infinite kurtosis.

The results firstly show that the bias of OLS is large. For the larger sample sizes sim-

$\tilde{x}_{r(v,t)}$ is not a linear combination of the other observations $x_{r(v,t)}$, $j \neq t$. In other words there is a $u \in \mathbb{R}^K$ satisfying $\tilde{x}'_{r(v,j)}u = 0$ for all $j \neq t$ but $\tilde{x}'_{r(v,t)}u \neq 0$. As a consequence, OLS regression always delivers a perfect fit for observation t: Suppose $y_{r(v,t)} - \tilde{x}'_{r(v,t)}b \neq 0$ for some b. Thus b cannot be the OLS estimator as there is a $c \in \mathbb{R}$ such that adding $c\,u$ to b sets the residual for observation t to zero ($y_{r(v,t)} - \tilde{x}'_{r(v,t)}(b + cu) = 0$) without affecting any of the other residuals. There can only be a perfect fit regardless of Y if $\tilde{x}_{r(v,t)}(\tilde{\mathbf{X}}'_v\tilde{\mathbf{X}}_v)^{-1}\tilde{x}_{r(v,t)} = 1$.

Table 1: Simulation results for $h = 1$

	Bias			**Std. dev.**		
Sample size	OLS	M&S	DaC	OLS	M&S	DaC
$\nu = \mathbf{2.1}$, $p = \mathbf{1}$						
33	-0.728	-0.100	-0.069	0.882	0.830	1.032
100	-0.776	-0.090	0.022	0.570	0.530	0.590
300	-0.748	-0.082	-0.001	0.363	0.344	0.359
900	-0.731	-0.085	0.005	0.281	0.241	0.227
2700	-0.711	-0.076	0.002	0.204	0.191	0.148
$\nu = \mathbf{2.1}$, $p = \mathbf{2}$						
33	-0.777	-0.192	0.015	0.965	0.972	1.162
100	-0.846	-0.226	-0.017	0.675	0.675	0.675
300	-0.842	-0.190	0.003	0.551	0.545	0.495
900	-0.832	-0.226	0.020	0.541	0.527	0.464
$\nu = \mathbf{2.5}$, $p = \mathbf{1}$						
33	-0.530	-0.096	-0.021	0.713	0.711	0.875
100	-0.537	-0.063	-0.011	0.444	0.445	0.460
300	-0.505	-0.035	-0.002	0.265	0.248	0.265
900	-0.478	-0.028	0.003	0.168	0.144	0.154
$\nu = \mathbf{2.5}$, $p = \mathbf{2}$						
33	-0.508	-0.135	-0.028	0.785	0.760	0.897
100	-0.544	-0.094	0.021	0.509	0.463	0.493
300	-0.540	-0.069	-0.011	0.353	0.319	0.304
900	-0.516	-0.092	-0.007	0.276	0.252	0.221

Notes: $K/T = 0.25$, $a_1 = 0.2$, $\beta = 0.5$, $\sigma = 1$, $\rho = 0.9$, $\sigma_\eta = 0.2$.

Table 2: Simulation results for $h = 3$

	Bias			**Std. dev.**		
Sample size	OLS	M&S	DaC	OLS	M&S	DaC
$\nu = \mathbf{2.1}$, $p = \mathbf{1}$						
33	-1.057	-0.316	-0.092	1.426	1.400	2.467
100	-1.016	-0.195	-0.026	0.851	0.815	1.250
300	-0.897	-0.144	-0.004	0.519	0.479	0.627
900	-0.874	-0.131	0.002	0.388	0.298	0.351
$\nu = \mathbf{2.1}$, $p = \mathbf{2}$						
33	-1.171	-0.469	-0.252	1.474	1.539	3.239
100	-1.179	-0.369	-0.003	0.992	0.976	1.199
300	-1.083	-0.345	0.021	0.761	0.723	0.634
900	-0.983	-0.388	0.005	0.614	0.585	0.375
$\nu = \mathbf{2.5}$, $p = \mathbf{1}$						
33	-0.522	-0.115	-0.017	0.923	0.916	1.562
100	-0.481	-0.081	0.037	0.546	0.534	0.923
300	-0.445	-0.026	0.008	0.313	0.300	0.447
900	-0.404	-0.026	0.001	0.187	0.171	0.237
$\nu = \mathbf{2.5}$, $p = \mathbf{2}$						
33	-0.530	-0.218	0.032	0.950	1.010	1.648
100	-0.540	-0.127	-0.004	0.608	0.592	0.824
300	-0.467	-0.099	0.010	0.371	0.333	0.438
900	-0.457	-0.078	-0.002	0.268	0.220	0.232

Notes: $K/T = 0.2$, $a_1 = 0.1$, $a_2 = 0.05$, $a_3 = 0.025$, $\beta = 0.5$, $\sigma = 1$, $\rho = 0.9$, $\sigma_\eta = 0.2$. Number of replications: 1000.

ulated, the bias of OLS exceeds its standard deviation by a mulitple. The bias of M&S is smaller but non-negligible relative to its standard deviation. It is especially pronounced when the regressors are more heavy-taield (smaller ν) and when the error variance is a convex function of $\|\tilde{x}_t\|$ ($p > 1$). Any loss in efficiency of the DaC estimator relative to M&S and OLS is barely visible in the one-period feedback case (h=1) and moderate in the three-period feedback case (h=3).

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A Proofs

I first introduce some useful notation: First, remember that T is assumed to be a multiple of $h+1$ and that the subsample size is denoted by $\bar{T} = T/(h+1)$. Let $\varepsilon = (\varepsilon_{1-h}, \cdots, \varepsilon_T)$, the vector of all error terms affecting observed regressors or outcomes. For $v \in \{1-h, \ldots, h+1\}$, define the $\bar{T} \times (T+h)$ selection matrix

$$\mathbf{S}_v = \begin{pmatrix} e'_{h+v} \\ e'_{v+2h+1} \\ e'_{v+3h+2} \\ \vdots \\ e'_{T-(h+1)+v} \end{pmatrix},$$

where $e_l \in \mathbb{R}^{T+h}$ is the lth canonical basis vector of $\mathbb{R}^{T+h}$. (A vector of length $T+h$ whose only non-zero element is the lth element, which is equal to 1.) It selects for $v \geq 1$ periods belonging of the vth subsample, and for $v \leq 0$ their lagged counterparts from a vector containing period $1-h$ through T. For example $\mathbf{S}_1\varepsilon = (\varepsilon_1, \varepsilon_{2+h}, \varepsilon_{3+2h}, \ldots, \varepsilon_{T-h})'$ and $\mathbf{S}_{-1}\varepsilon = (\varepsilon_{-1}, \varepsilon_h, \varepsilon_{1+2h}, \ldots, \varepsilon_{T-h-2})'$. Let b be any $(T+h)$-vector. Note that for values of $v, q \in \{1-h, \ldots, h+1\}$:

$$\mathbf{S}_v diag(b) \mathbf{S}'_q = \begin{cases} diag(\mathbf{S}_v b) & \text{if } v = q \\ \begin{pmatrix} \mathbf{0}_{\bar{T}-1} & diag(\sigma_v^2, \sigma_{v+h+1}^2, \ldots, \sigma_{T-2(h+1)+v}^2) \\ 0 & \mathbf{0}'_{\bar{T}-1} \end{pmatrix} & \text{if } v \geq 1, q = v-(h+1) \\ \begin{pmatrix} \mathbf{0}'_{\bar{T}-1} & 0 \\ diag(\sigma_v^2, \sigma_{v+h+1}^2, \ldots, \sigma_{T-2(h+1)+v}^2) & \mathbf{0}_{\bar{T}-1} \end{pmatrix} & \text{if } v \leq 0, q = v+h+1 \\ \mathbf{0}_{\bar{T}\times\bar{T}} & \text{otherwise.} \end{cases}$$

and that $\mathbf{S}'_v\mathbf{S}_v$ is a diagonal matrix whose only non-zero element is 1, whence $\|\mathbf{S}_v\|_2 = 1$. Define the matrix of regressors from subsample v:

$$\mathbf{X}_v = \begin{pmatrix} x'_v \\ x'_{v+h+1} \\ x'_{v+2h+2} \\ \vdots \end{pmatrix} \quad \text{and} \quad \tilde{\mathbf{X}}_v = \begin{pmatrix} \tilde{x}'_v \\ \tilde{x}'_{v+h+1} \\ \tilde{x}'_{v+2h+2} \\ \vdots \end{pmatrix}$$

and $Y_v = (y_v, y_{v+h+1}, y_{v+2h+2}, \dots)'$ the vectors of outcomes from subsample v. Let $\boldsymbol{\Xi} = (\alpha_1, \dots, \alpha_h)$ and $U_v = (\mathbf{S}_{v-1}\varepsilon, \mathbf{S}_{v-2}\varepsilon, \dots, \mathbf{S}_{v-h}\varepsilon)$. Note that under assumption *Feedback Decomposition*, $\mathbf{X}_v$ can be written as

$$\mathbf{X}_v = \tilde{\mathbf{X}}_v + \sum_{l=1}^{h} \mathbf{S}_{v-l}\varepsilon\alpha_l' = \tilde{\mathbf{X}}_v + U_v\boldsymbol{\Xi}'$$

The following lemma is a version of Lemma 5 from Mikusheva and Sølvsten (2025). It will be used to transform the regressors in such a way that only a fixed number of them are affected by feedback while the remaining regressors are strictly exogenous and that $r'\beta$ depends only on a fixed number of parameters.

Lemma 1. *For $K \geq h+1$, let the $(h+1) \times K$ matrix $\mathbf{A}$ be given by $\mathbf{A} = (r, \alpha_1, \dots, \alpha_h)'$. For each subsample $v \in \{1, \dots, h+1\}$, there is an invertible $K \times K$ matrix $\boldsymbol{\Theta}$ satisfying:*

- $\boldsymbol{\Theta}'\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v\boldsymbol{\Theta} = \bar{T}\mathbf{I}_K$,
- $\mathbf{A}\boldsymbol{\Theta} = \begin{bmatrix}\mathbf{A}^{\star\prime} & \mathbf{0}_{(h+1)\times(K-h-1)}\end{bmatrix}$ *where* $\mathbf{A}^\star \in \mathbb{R}^{(h+1)\times(h+1)}$ *satisfies* $\mathbf{A}^{\star\prime}\mathbf{A}^\star = \bar{T}\mathbf{A}(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v)^{-1}\mathbf{A}'$ *and* $\mathbf{0}_{(h+1)\times(K-h-1)}$ *is a $(h+1) \times (K-h-1)$ matrix of zeros,*
- *and* $\boldsymbol{\Theta}\boldsymbol{\Theta}' = \left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}\right)^{-1} = \bar{T}\left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v\right)^{-1}$*.. Thus, the eigenvalues of $\boldsymbol{\Theta}\boldsymbol{\Theta}'$ are those of $(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T})^{-1}$, too.*

Proof. We construct $\boldsymbol{\Theta}$ as follows: Let be $\boldsymbol{\Sigma}$ is the symmetric square root of $\frac{1}{\bar{T}}\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v$. Let $\mathbf{Q}\mathbf{A}^\star$ be the QR-decomposition of $\boldsymbol{\Sigma}^{-1}\mathbf{A}'$, meaning $\mathbf{Q}$ is a $K \times (h+1)$ matrix whose columns are orthonormal, $\mathbf{A}^\star$ is an upper triangular $(h+1) \times (h+1)$ matrix and $\mathbf{Q}\mathbf{A}^\star = \boldsymbol{\Sigma}^{-1}\mathbf{A}'$. Let $v_1, v_2, ...v_{K-h-1} \in \mathbb{R}^K$ be any orthonormal set of $K-h-1$ vectors, each of which are orthogonal to all columns of $\mathbf{Q}$. Let $\boldsymbol{\Gamma} = [\mathbf{Q}, v_1, \dots, v_{K-h-1}]$ (note $\boldsymbol{\Gamma}' = \boldsymbol{\Gamma}^{-1}$) and $\bar{\mathbf{A}} = \begin{pmatrix}\mathbf{A}^{\star\prime} & \mathbf{0}_{(h+1)\times(K-h-1)}\end{pmatrix}'$. Note that $\boldsymbol{\Gamma}\bar{\mathbf{A}} = \mathbf{Q}\mathbf{A}^\star = \boldsymbol{\Sigma}^{-1}\mathbf{A}'$. Then $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\Gamma}$. We verify the properties stated in the lemma:

$$\boldsymbol{\Theta}'\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v\boldsymbol{\Theta} = \boldsymbol{\Gamma}'\boldsymbol{\Sigma}^{-1}\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v\boldsymbol{\Sigma}^{-1}\boldsymbol{\Gamma} = \bar{T}\boldsymbol{\Gamma}'\mathbf{I}_K\boldsymbol{\Gamma} = \bar{T}\mathbf{I}_K,$$

$$\mathbf{A}\boldsymbol{\Theta} = \mathbf{A}\boldsymbol{\Sigma}^{-1}\boldsymbol{\Gamma} = \bar{\mathbf{A}}'\boldsymbol{\Gamma}'\boldsymbol{\Gamma} = \bar{\mathbf{A}}' = \begin{pmatrix}\mathbf{A}^{\star\prime} & \mathbf{0}_{(h+1)\times(K-h-1)}\end{pmatrix}$$

$$\mathbf{A}^{\star\prime}\mathbf{A}^\star = \bar{\mathbf{A}}'\bar{\mathbf{A}} = \mathbf{A}\boldsymbol{\Sigma}^{-1}\boldsymbol{\Gamma}\boldsymbol{\Gamma}'\boldsymbol{\Sigma}^{-1}\mathbf{A}' = \bar{T}\mathbf{A}(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v)^{-1}\mathbf{A}'$$

Finally, since $\boldsymbol{\Gamma}$ is orthogonal ($\boldsymbol{\Gamma}\boldsymbol{\Gamma}' = \mathbf{I}_K$) and $\boldsymbol{\Sigma}^{-1}$ is symmetric,

$$\boldsymbol{\Theta}\boldsymbol{\Theta}' = \boldsymbol{\Sigma}^{-1}\boldsymbol{\Gamma}\boldsymbol{\Gamma}'\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}^{-2} = \left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v/\bar{T}\right)^{-1} = \bar{T}\left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v\right)^{-1}.$$

As $\boldsymbol{\Theta}$ is square, the eigenvalues of $\boldsymbol{\Theta}'\boldsymbol{\Theta}$ are exactly those of $\bar{T}\left(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v\right)^{-1}$. □

The following lemma is a version of Lemma 7 form Mikusheva and Sølvsten (2025) with a slight change:

Lemma 2. *Suppose* $\tilde{\mathbf{X}}$ *is a* $T \times (h+1)$ *matrix with* $\tilde{\mathbf{X}}'\tilde{\mathbf{X}}/T = \mathbf{I}_{h+1}$, $\mathbf{B}$ *is a* $T \times T$ *matrix,* $\alpha_1, \ldots, \alpha_h$ *are non-random* $h+1$*- vectors, and* $u_1, \ldots, u_h$ *are independent vectors of dimension* T *of independent mean-zero random variables with variances* $0 < c \leq Var\left(u_{li}\right) \leq C < \infty$ *and* $\mathbb{E}\left[u_{li}^4\right] < C^2 < \infty$ *for all* l, i*. Then:*

1. $\left\| T^{-1/2}\tilde{\mathbf{X}}'\mathbf{B}\sum_{\ell=0}^{h} u_\ell \alpha_\ell' \right\|_F = O_p(\max_l \|\alpha_l\| \|\mathbf{B}\|_2)$,
2. $u_1'\mathbf{B}u_1 - \mathbb{E}[u_1'\mathbf{B}u_1] = O_p(\|\mathbf{B}\|_F)$.

Proof. 1. Note that:

$$\begin{aligned}
\mathbb{E}\left\| \frac{1}{\sqrt{T}}\tilde{\mathbf{X}}'\mathbf{B}\sum_{\ell=0}^{h} u_\ell \alpha_\ell' \right\|_F^2 &= \operatorname{tr}\left[\frac{\tilde{\mathbf{X}}'\tilde{\mathbf{X}}}{T}\mathbf{B}\sum_{\ell,j=1}^{h} \mathbb{E}[u_\ell u_j']\alpha_\ell'\alpha_j\mathbf{B}' \right] \\
&\leq \operatorname{tr}\left[\frac{\tilde{\mathbf{X}}'\tilde{\mathbf{X}}}{T} \right] \left\| \mathbf{B}\sum_{\ell=1}^{h} \mathbb{E}[u_\ell u_\ell']\|\alpha_\ell\|^2\mathbf{B}' \right\|_2 \\
&\leq h\|\mathbf{B}\|_2^2 \left(\sum_{\ell=1}^{h} \|\alpha_\ell\|^2 \cdot \|\mathbb{E}[u_\ell u_\ell']\|_2 \right) \\
&\leq Ch^2 \max \|\alpha_\ell\|^2 \|\mathbf{B}\|_2^2.
\end{aligned}$$

The last inequality uses that the $\mathbb{E}\left[u_l u_l'\right]$ are diagonal matrices whose largest elements are at most C. Then Markov's inequality leads to the result.

2. We have:

$$\begin{aligned}
u_1'\mathbf{B}u_1 - \mathbb{E}[u_1'\mathbf{B}u_1] &= \sum_t \sum_{s \neq t} \mathbf{B}_{ts} u_{1t} u_{1s} + \sum_t \mathbf{B}_{tt}(u_{1t}^2 - \sigma^2) \\
&= \sum_{t>s} (\mathbf{B}_{ts} + \mathbf{B}_{st}) u_{1s} u_{1t} + \sum_t \mathbf{B}_{tt}(u_{1t}^2 - \sigma^2)
\end{aligned}$$

In the summations above, the summands are uncorrelated. Therefore,

$$\begin{aligned}
\mathbb{E}\left[(u_1'\mathbf{B}u_1 - \mathbb{E}[u_1'\mathbf{B}u_1])^2 \right] &= \sum_{t>s} (\mathbf{B}_{ts} + \mathbf{B}_{ts})^2 \sigma_s^2 \sigma_t^2 + \sum_t \mathbf{B}_{tt}^2 \mathbb{E}[(u_{1t}^2 - \sigma^2)^2] \\
&\leq C^2 \left(\sum_{t>s} (\mathbf{B}_{ts} + \mathbf{B}_{ts})^2 + \sum_t \mathbf{B}_{tt}^2 \right) \\
&\leq C^2 \frac{1}{2} \|\mathbf{B} + \mathbf{B}'\|_F^2 \\
&= O(\|\mathbf{B}\|_F^2)
\end{aligned}$$

The result is then implied by Markov's inequality.

□

Definition 2. For each subsample v, let $\mathbf{\Theta}_v$ be the transformation from lemma 1, let $\mathbf{\Theta}_{v,1}$ denote its first $h+1$ columns and $\mathbf{\Theta}_{v,2}$ the remaining columns. Let

- $\mathbf{Z}_{1,v} = \mathbf{X}_v\mathbf{\Theta}_{v,1}, \mathbf{Z}_{2,v} = \mathbf{X}_v\mathbf{\Theta}_{v,2}, \tilde{\mathbf{Z}}_{1,v} = \tilde{\mathbf{X}}_v\mathbf{\Theta}_{v,1}$
- $\mathbf{M}_v = \mathbf{I}_{\bar{T}} - \mathbf{Z}_{2,v}(\mathbf{Z}'_{2,v}\mathbf{Z}_{2,v})^{-1}\mathbf{Z}'_{2,v}, \mathbf{Q}_v = \mathbb{E}\left[\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right]$
- $z_{1,t} = \mathbf{\Theta}'_{u(t),1}x_t, z_{2,t} = \mathbf{\Theta}'_{u(t),2}x_t, \tilde{z}_{1,t} = \mathbf{\Theta}'_{u(t),1}\tilde{x}_t$.

Define the $h+1$ vectors $r_v, \alpha_{v,1}, \ldots, \alpha_{v,h} \in \mathbb{R}^{h+1}$ by $r'\mathbf{\Theta}_v = (r_v{}', 0'_{K-h-1})$ and $\alpha'_l\mathbf{\Theta}_v = (\alpha'_{v,l}, 0'_{K-h-1})$.

Lemma 3 (Eigenvalue lower bound for the partialled block)**.** *Under assumptions* Feedback Decomposition*,* Bounded Eigenvalues*,* Regular Errors *and* Moderate-Dimensional Regime*, for every subsample* $v \in \{1, \ldots, h+1\}$*,*

$$\lambda_{\min}\Big(\frac{1}{\bar{T}}\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\Big) \;\geq\; 1 - o_p(1).$$

In particular, $\|(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T})^{-1}\|_2 = O_p(1)$.

Proof. From Lemma 1:

$$\mathbf{M}_v\widetilde{\mathbf{Z}}_{1,v} = \widetilde{\mathbf{Z}}_{1,v}, \qquad \frac{1}{\bar{T}}\widetilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v\widetilde{\mathbf{Z}}_{1,v} = \frac{1}{\bar{T}}\widetilde{\mathbf{Z}}'_{1,v}\widetilde{\mathbf{Z}}_{1,v} = \mathbf{I}_{h+1}. \tag{1}$$

Hence

$$\frac{1}{\bar{T}}\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v} = \mathbf{I}_{h+1} + \mathbf{\Delta}_v^{(1)} + \mathbf{\Delta}_v^{(2)} + \mathbf{\Delta}_v^{(3)}, \tag{2}$$

where

$$\mathbf{\Delta}_v^{(1)} := \frac{1}{\bar{T}}\widetilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v\sum_{\ell=1}^{h}\mathbf{S}_{v-\ell}\varepsilon\,\alpha'_{v,\ell} + \left(\frac{1}{\bar{T}}\widetilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v\sum_{\ell=1}^{h}\mathbf{S}_{v-\ell}\varepsilon\,\alpha'_{v,\ell}\right)', \tag{3}$$

$$\mathbf{\Delta}_v^{(2)} := \sum_{\ell=1}^{h}\alpha_{v,\ell}\alpha'_{v,\ell}\frac{1}{\bar{T}}\sum_{s=1}^{\bar{T}}\sigma^2_{v-\ell+(s-1)(h+1)}M_{v,ss}, \tag{4}$$

$$\mathbf{\Delta}_v^{(3)} := \frac{1}{\bar{T}}\sum_{\ell,k=1}^{h}\alpha_{v,\ell}\alpha'_{v,k}\Big(\varepsilon'\mathbf{S}'_{v-\ell}\mathbf{M}_v\mathbf{S}_{v-k}\varepsilon - \mathbb{E}\big[\varepsilon'\mathbf{S}'_{v-\ell}\mathbf{M}_v\mathbf{S}_{v-k}\varepsilon\big]\Big). \tag{5}$$

The decomposition (2) separates the deterministic part $\mathbf{I}_{h+1} + \mathbf{\Delta}_v^{(2)}$ from the random perturbations $\mathbf{\Delta}_v^{(1)}$ and $\mathbf{\Delta}_v^{(3)}$. We apply Weyl's inequality sequentially. Since all matrices in (2) are symmetric,

$$\lambda_{\min}\Big(\frac{1}{\bar{T}}\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\Big) \geq \lambda_{\min}(\mathbf{I}_{h+1} + \mathbf{\Delta}_v^{(2)}) + \lambda_{\min}(\mathbf{\Delta}_v^{(1)}) + \lambda_{\min}(\mathbf{\Delta}_v^{(3)}). \tag{6}$$

Step 1 (lower bound on $\mathbf{I}_{h+1} + \mathbf{\Delta}_v^{(2)}$). The diagonal elements of $\mathbf{M}_v$ satisfy $M_{v,ss} \geq 0$ for all s (since $\mathbf{M}_v$ is an orthogonal projection), so each rank-one term $\alpha_{v,\ell}\alpha'_{v,\ell'}\cdot\frac{1}{\bar{T}}\sum_s \sigma^2_{v-\ell+(s-1)(h+1)}M_{v,ss}$ is positive semidefinite. Therefore $\mathbf{\Delta}_v^{(2)} \succeq 0$ and

$$\lambda_{\min}(\mathbf{I}_{h+1} + \mathbf{\Delta}_v^{(2)}) \geq 1. \tag{7}$$

Step 2 (rate of $\boldsymbol{\Delta}_v^{(1)}$). For any symmetric matrix $\mathbf{B}$, $\lambda_{\min}(\mathbf{B}) \geq -\|\mathbf{B}\|_2$. Applying Lemma 2 with $\widetilde{\mathbf{X}} \equiv \widetilde{\mathbf{Z}}_{1,v}$ (satisfying $\widetilde{\mathbf{Z}}'_{1,v}\widetilde{\mathbf{Z}}_{1,v}/\bar{T} = \mathbf{I}_{h+1}$ by (1)), $\mathbf{B} \equiv \mathbf{M}_v$, and noting $\|\alpha_{v,\ell}\| \leq C$ under Assumption *Bounded Eigenvalues*,

$$\|\boldsymbol{\Delta}_v^{(1)}\|_2 \leq 2\Big\|\frac{1}{\bar{T}}\widetilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v \sum_{\ell} \mathbf{S}_{v-\ell\varepsilon}\,\alpha'_{v,\ell}\Big\|_F = O_p\Big(\frac{\|\mathbf{M}_v\|_2}{\sqrt{\bar{T}}}\Big) = O_p\big(\bar{T}^{-1/2}\big), \tag{8}$$

so $\lambda_{\min}(\boldsymbol{\Delta}_v^{(1)}) \geq -\|\boldsymbol{\Delta}_v^{(1)}\|_2 = O_p(\bar{T}^{-1/2}) = o_p(1)$.

Step 3 (rate of $\boldsymbol{\Delta}_v^{(3)}$). Again $\lambda_{\min}(\boldsymbol{\Delta}_v^{(3)}) \geq -\|\boldsymbol{\Delta}_v^{(3)}\|_2 \geq -\|\boldsymbol{\Delta}_v^{(3)}\|_F$. We bound the Frobenius norm term by term. Fix $\ell, k \in \{1, \ldots, h\}$ and set $u = \mathbf{S}_{v-\ell\varepsilon}\varepsilon$, $w = \mathbf{S}_{v-k\varepsilon}\varepsilon$. We bound the centered scalar $u'\mathbf{M}_v w - \mathbb{E}[u'\mathbf{M}_v w]$ in each case.

Case $\ell = k$. Here $u = w$ and $u'\mathbf{M}_v u - \mathbb{E}[u'\mathbf{M}_v u]$ is a centered quadratic form in independent variables. By Lemma 2,

$$u'\mathbf{M}_v u - \mathbb{E}[u'\mathbf{M}_v u] = O_p(\|\mathbf{M}_v\|_F) = O_p(\bar{T}^{1/2}).$$

Case $\ell \neq k$. Here u, w are independent, so $\mathbb{E}[u'\mathbf{M}_v w] = 0$, and the summands $M_{v,ts}u_t w_s$ are pairwise uncorrelated. Hence

$$\mathrm{Var}(u'\mathbf{M}_v w) = \sum_{t,s} M_{v,ts}^2 \,\mathrm{Var}(u_t)\,\mathrm{Var}(w_s) \leq C^2 \sum_{t,s} M_{v,ts}^2 = C^2\|\mathbf{M}_v\|_F^2,$$

so by Chebyshev's inequality $u'\mathbf{M}_v w = O_p(\|\mathbf{M}_v\|_F) = O_p(\bar{T}^{1/2})$.

In both cases the centered form is $O_p(\bar{T}^{1/2})$. With $\|\alpha_{v,\ell}\alpha'_{v,k}\|_F = \|\alpha_{v,\ell}\|\|\alpha_{v,k}\| \leq C^2$ and a fixed double sum over h^2 pairs,

$$\|\boldsymbol{\Delta}_v^{(3)}\|_F \leq \frac{1}{\bar{T}} \sum_{\ell,k=1}^{h} \|\alpha_{v,\ell}\alpha'_{v,k}\|_F \cdot O_p(\|\mathbf{M}_v\|_F) = \frac{h^2 C^2 \cdot O_p(\bar{T}^{1/2})}{\bar{T}} = O_p\big(\bar{T}^{-1/2}\big), \tag{9}$$

so $\lambda_{\min}(\boldsymbol{\Delta}_v^{(3)}) \geq -\|\boldsymbol{\Delta}_v^{(3)}\|_F = O_p(\bar{T}^{-1/2}) = o_p(1)$.

Conclusion. Substituting (7), (8), and (9) into (6),

$$\lambda_{\min}\Big(\frac{1}{\bar{T}}\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\Big) \;\geq\; 1 + 0 - o_p(1) - o_p(1) \;=\; 1 - o_p(1).$$

For any fixed $\delta \in (0,1)$ it follows that $\Pr[\lambda_{\min}(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T}) \geq 1-\delta] \to 1$, so $(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T})^{-1}$ exists with probability approaching 1 and $\|(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T})^{-1}\|_2 \leq (1-\delta)^{-1} = O_p(1)$. □

Lemma 4. *For each subsample $v \in \{1, \ldots, h+1\}$: $\lambda_{\min}(\mathbf{Q}_v) = \Omega(T)$, $\lambda_{\max}(\mathbf{Q}_v) = O(T)$.*

Proof. Using

$$\mathbf{Q}_v/T = \mathbf{I}_{h+1} + \sum_{l} \alpha_{v,l}\alpha'_{v,l} \sum_{t=1}^{\bar{T}} \sigma^2_{r(v,t)}\mathbf{M}_{v,tt},$$

we use Weyl's inequality to establish that the smallest eigenvalue of $\mathbf{Q}_v/\bar{T}$ is bounded below:

$$\lambda_{\min}\left(\mathbf{Q}_v/\bar{T}\right) \geq \lambda_{\min}\left(\mathbf{I}_{h+1}\right) + \lambda_{\min}\left(\sum_{l=1}^{h+1} \alpha_{v,l}\alpha'_{v,l}\frac{1}{\bar{T}}\sum_{s=1}^{\bar{T}} \sigma^2_{v-l+(s-1)(h+1)}\mathbf{M}_{v,s+1,s+1}\right) \geq 1$$

The second equality follows from the fact that the diagonal elements of an orthogonal projection matrix are non-negative making the entire second matrix positive semi-definite. To bound $\lambda_{\max}\left(\mathbf{Q}_v/\bar{T}\right)$ above, we use the triangle inequality:

$$\begin{aligned}\lambda_{\max}\left(\mathbf{Q}_v/\bar{T}\right) \leq & \lambda_{\max}\left(\mathbf{I}_{h+1}\right) + \frac{1}{\bar{\bar{T}}}\sum_{l=1}^{h+1}\|\alpha_{v,l}\|^2 \sum_{s=1}^{\bar{T}} \sigma^2_{v-l+(s-1)(h+1)}\mathbf{M}_{v,s+1,s+1} \\ \leq & 1 + \frac{C}{\bar{\bar{T}}}\sum_{l=1}^{h+1}\|\alpha_{v,l}\|^2(\bar{T}-K) \\ \leq & 1 + C^3(h+1)\end{aligned}$$

□

Lemma 5. *Under assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors *and* Moderate-Dimensional Regime,

$$\left\|\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} - \mathbf{Q}_v^{-1}\right\|_2 = O_p(\bar{T}^{-3/2})$$

Proof. We will use the identity

$$A^{-1} - B^{-1} = A^{-1}(B-A)B^{-1}$$

(where A and B are any two invertible matrices of the same dimension), which yields

$$\left\|A^{-1} - B^{-1}\right\|_2 \leq \left\|A^{-1}\right\|_2 \left\|B^{-1}\right\|_2 \|B-A\|_2$$

Applying this with $A = \mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T}$ and $B = \mathbf{Q}_v/\bar{T}$:

$$\begin{aligned}\frac{1}{\bar{\bar{T}}}\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v} - \mathbf{Q}_v\right) = & \frac{1}{\bar{\bar{T}}}\tilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v\sum_{l=1}^{h}\mathbf{S}_{v-l}\varepsilon\alpha'_{v,l} + \frac{1}{\bar{\bar{T}}}\sum_{l=1}^{h}\alpha_{v,l}\varepsilon'\mathbf{S}'_{v-l}\mathbf{M}_v\tilde{\mathbf{Z}}_{1,v} + \\ & \frac{1}{\bar{\bar{T}}}\sum_{l,k=1}^{h}\alpha_{v,l}\alpha'_{v,k}\left(\varepsilon'\mathbf{S}'_{v-l}\mathbf{M}_v\mathbf{S}'_{v-k}\varepsilon - \mathbb{E}\left[\varepsilon'\mathbf{S}'_{v-l}\mathbf{M}_v\mathbf{S}'_{v-k}\varepsilon\right]\right)\end{aligned} \tag{10}$$

Given Lemma 1, $\|\alpha_{v,l}\|^2 = \bar{T}\alpha'_l\left(\tilde{\mathbf{X}}'_v\tilde{\mathbf{X}}_v\right)^{-1}\alpha_l$, which is bounded for each v, l by assumption *Bounded Eigenvalues*. Therefore applying Lemma 2 to these terms yields:

$$\left\|\frac{1}{\bar{\bar{T}}}\tilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v\sum_{l=1}^{h}\mathbf{S}_{v-l}\varepsilon\alpha'_{v,l}\right\|_F = O_p\left(\frac{1}{\sqrt{\bar{\bar{T}}}}\|\mathbf{M}_v\|_2\right) = O_p\left(\frac{1}{\sqrt{\bar{\bar{T}}}}\right)$$

and

$$\begin{aligned}\frac{1}{\bar{\bar{T}}}\left(\varepsilon'\mathbf{S}'_{v-l}\mathbf{M}_v\mathbf{S}_{v-k}\varepsilon - \mathbb{E}\left[\varepsilon'\mathbf{S}'_{v-l}\mathbf{M}_v\mathbf{S}_{v-k}\varepsilon\right]\right) = & O_p\left(\frac{1}{\bar{\bar{T}}}\left\|\mathbf{S}'_{v-l}\mathbf{M}_v\mathbf{S}_{v-k}\right\|_F\right) \\ = & O_p\left(\frac{1}{\bar{\bar{T}}}\|\mathbf{M}_v\|_F\right) = O_p\left(\frac{1}{\sqrt{\bar{\bar{T}}}}\right)\end{aligned}$$

Here we used that $\|\mathbf{S}_a'\mathbf{M}_b\mathbf{S}_c\|_F = \|\mathbf{M}_b\|_F$ ($\mathbf{S}_a'\mathbf{M}_b\mathbf{S}_c$ is a spread-out version of $\mathbf{M}_v$, i.e. each non-zero element corresponds to a distinct entry in $\mathbf{M}_v$) and $\|\mathbf{M}_v\|_F = \sqrt{\bar{T}-K}$. And since $\left\|\alpha_{v,l}\alpha_{v,k}'\right\|_F = \sqrt{\text{tr}\left(\alpha_{v,l}\alpha_{v,k}'\alpha_{v,k}\alpha_{v,l}'\right)} = \|\alpha_{v,l}\|\|\alpha_{v,k}\|$ is bounded, it follows that

$$\left\|\frac{1}{\bar{T}}\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v} - \mathbf{Q}_v\right)\right\|_2 \leq \left\|\frac{1}{\bar{T}}\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v} - \mathbf{Q}_v\right)\right\|_F = O_p(\bar{T}^{-\frac{1}{2}})$$

We know from Lemma 4 that $\lambda_{\min}\left(\mathbf{Q}_v/\bar{T}\right)$ is bounded below by a positive constant which implies that $\left\|(\mathbf{Q}_v/\bar{T})^{-1}\right\|_2 = \lambda_{\max}\left((\mathbf{Q}_v/\bar{T})^{-1}\right)$ is bounded above. Regarding $\left\|\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T}\right)^{-1}\right\|_2$, Lemma 3 implies that it is smaller than any $C > 1$ with probability approaching 1 implying that it is $O_p(1)$. □

Lemma 6. *Under assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors *and* Moderate-Dimensional Regime,

$$r'(\hat{\beta} - \beta) = \frac{1}{h+1}\sum_{v=1}^{h+1} r_v'\mathbf{Q}_v{}^{-1}\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon + O_p(1/\bar{T})$$

Proof. Applying the transformation $\mathbf{\Theta}_v$ to $\mathbf{X}_v$ yields:

$$\mathbf{Z}_{1,v} = \tilde{\mathbf{X}}_v\mathbf{\Theta}_{v,1} + \sum_{l=1}^{h}\mathbf{S}_{v-l}\varepsilon\alpha_{v,l}'$$

and $\mathbf{Z}_{2,v} = \tilde{\mathbf{X}}_v\mathbf{\Theta}_2$ is strictly exogenous as $\mathbf{\Xi}'\mathbf{\Theta}_2 = \mathbf{0}_{h,K-h-1}$. Let $\mathbf{M}_v = \mathbf{I}_{\bar{T}} - \mathbf{P}_v$ where $\mathbf{P}_v = \mathbf{Z}_{2,v}(\mathbf{Z}_{2,v}'\mathbf{Z}_{2,v})^{-1}\mathbf{Z}_{2,v}'$.

$$\begin{aligned}
(h+1)r'(\hat{\beta} - \beta) &= \sum_{v=1}^{h+1} r'(\mathbf{X}_v'\mathbf{X}_v)^{-1}\mathbf{X}_v'\mathbf{S}_v\varepsilon \\
&= \sum_{v=1}^{h+1} r'\mathbf{\Theta}_v(\mathbf{Z}_v'\mathbf{Z}_v)^{-1}\mathbf{Z}_v'\mathbf{S}_v\varepsilon \\
&= \sum_{v=1}^{h+1} r_v'\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1}\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon
\end{aligned}$$

The last equality uses the Frisch-Waugh-Lovell theorem. Let $\mathbf{M}_{vs,s}$ denote the diagonal elements of $\mathbf{M}_v$. Define

$$\begin{aligned}
\mathbf{Q}_v =& \mathbb{E}\left[\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right] \\
&= \left(\tilde{\mathbf{Z}}_{1,v}'\tilde{\mathbf{Z}}_{1,v} + \sum_{l=1}^{h}\alpha_{v,l}\alpha_{v,l}'\sum_{s=1}^{\bar{T}}\sigma^2_{v-l+(s-1)(h+1)}\mathbf{M}_{vs,s}\right) \\
&= \left(\bar{T}\mathbf{I}_{h+1} + \sum_{l=1}^{h}\alpha_{v,l}\alpha_{v,l}'\sum_{s=1}^{\bar{T}}\sigma^2_{v-l+(s-1)(h+1)}\mathbf{M}_{vs,s}\right)
\end{aligned}$$

We can decompose the quantity of interest as follows:

$$(h+1)r'(\hat{\beta}_h - \beta) = \sum_{v=1}^{h+1} r_v' \mathbf{Q}_v{}^{-1} \mathbf{Z}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon + \sum_{v=1}^{h+1} r_v' \Delta \frac{1}{\bar{T}} \mathbf{Z}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon \tag{11}$$

where $\Delta := \left(\mathbf{Z}_{1,v}' \mathbf{M}_v \mathbf{Z}_{1,v}/\bar{T}\right)^{-1} - \left(\mathbf{Q}_v/\bar{T}\right)^{-1}$. We have to show the second term of this expression to be $O_p(\bar{T}^{-1})$. W know from Lemma 5 that $\|\Delta\|_2 = O(\bar{T}^{-1/2})$.

It remains to bound

$$\frac{1}{\bar{T}} \mathbf{Z}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon = \frac{1}{\bar{T}} \left(\tilde{\mathbf{Z}}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon + \sum_{l=1}^{h} \alpha_{v,l} \varepsilon \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \varepsilon \right)$$

Lemma 2 can be applied to these terms: Part (i) with $\alpha_1 = (1, 0, 0, \dots)'$ and $\alpha_2 = \cdots = \alpha_{h+1} = 0$ can be applied to $\frac{1}{\bar{T}} \left(\tilde{\mathbf{Z}}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon \right)$ to show that

$$\left\| \frac{1}{\bar{T}} \left(\tilde{\mathbf{Z}}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon \right) \right\| = O_p \left(\frac{1}{\sqrt{\bar{T}}} \|\mathbf{M}_v\|_2 \right) = O_p \left(\frac{1}{\sqrt{\bar{T}}} \right).$$

Part (ii) can be applied to each term $\varepsilon \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \varepsilon$ given that its expectation is equal to zero. Together with the fact that the $\|\alpha_{v,l}\|$ are bounded, we obtain:

$$\left\| \frac{1}{\bar{T}} \alpha_{v,l} \varepsilon \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \varepsilon \right\| = O_p \left(\frac{1}{\bar{T}} \left\| \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \right\|_F \right) = O_p \left(\frac{1}{\sqrt{\bar{T}}} \right)$$

In conclusion, since $\|r_v\|$ is bounded, the second term of expression 11 is $O_p(1/\bar{T})$, which proves the lemma. □

Lemma 7. *Under Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, *and* Moderate-Dimensional Regime $\Sigma := Var\left(\frac{1}{h+1} \sum_{v=1}^{h+1} r_v' Q_v^{-1} Z_{1,v}' M_v S_v \varepsilon \right) \asymp T^{-1}$.

Proof. In the proof of Theorem 2, it is shown that Σ can be written as

$$(h+1)^2 \Sigma = \sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v} \mathbf{M}_v \mathbf{\Omega}_v^2 \mathbf{M}_v \tilde{\mathbf{Z}}_{1,v} \mathbf{Q}_v^{-1} r_v + \frac{1}{2} \|\mathbf{B} + \mathbf{B}'\|_F^2 \tag{12}$$

for some matrix $\mathbf{B}$ defined in the proof of Theorem 2. Turning to the first term, we first drop the $\mathbf{M}_v$ because $\tilde{\mathbf{Z}}_{1.v}' \tilde{\mathbf{Z}}_{2.v} = 0$. We then bound the expression in a standard way:

$$\begin{aligned} \sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v} \mathbf{M}_v \mathbf{\Omega}_v^2 \mathbf{M}_v \tilde{\mathbf{Z}}_{1,v} \mathbf{Q}_v^{-1} r_v &= \frac{1}{\bar{T}} \sum_v r_v' \bar{T} \mathbf{Q}_v^{-1} \frac{1}{\bar{T}} \tilde{\mathbf{Z}}_{1,v} \mathbf{\Omega}_v^2 \tilde{\mathbf{Z}}_{1,v} \bar{T} \mathbf{Q}_v^{-1} r_v \\ &\leq \frac{1}{\bar{T}} \sum_v \|r_v\|^2 \lambda_{\max} \left(T \mathbf{Q}_v^{-1} \right)^2 \lambda_{\max} \left(\frac{1}{\bar{T}} \tilde{\mathbf{Z}}_{1,v} \mathbf{\Omega}_v^2 \tilde{\mathbf{Z}}_{1,v} \right) \end{aligned}$$

We know from Lemma 4 that $\lambda_{\max}\left(T\mathbf{Q}_v^{-1}\right)$ is bounded. As $\tilde{\mathbf{Z}}_{1,v}/\sqrt{T}$ is orthonormal, $\lambda_{\max}\left(\frac{1}{T}\tilde{\mathbf{Z}}_{1,v}\mathbf{\Omega}_v^2\tilde{\mathbf{Z}}_{1,v}\right) = \lambda_{\max}\left(\mathbf{\Omega}_v^2\right) < C^2$. Finally, from Lemma 1, $\|r_v\| \leq C^2$. In conclusion, the first term of equation 12 is $O(T^{-1})$.

For the second term, $\frac{1}{2}\left\|\mathbf{B}+\mathbf{B}'\right\|_F^2 = \operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right) + \operatorname{tr}\left(\mathbf{B}^2\right)$. Using the Cauchy–Schwarz inequality for the trace inner product ($|\operatorname{tr}\left(A'C\right)| \leq \|A\|_F\|C\|_F$, applied with $A = \mathbf{B}'$ and $C = \mathbf{B}$):

$$|\operatorname{tr}\left(\mathbf{B}^2\right)| = |\operatorname{tr}\left((\mathbf{B}')'\mathbf{B}\right)| \leq \|\mathbf{B}'\|_F\|\mathbf{B}\|_F = \|\mathbf{B}\|_F^2 = \operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right), \tag{13}$$

using that transposition leaves the Frobenius norm unchanged and that $\operatorname{tr}\left(()\,\mathbf{B}\mathbf{B}'\right) = \operatorname{tr}\left(()\,\mathbf{B}'\mathbf{B}\right) = \|\mathbf{B}\|_F^2$ for any square matrix. Hence

$$\tfrac{1}{2}\left\|\mathbf{B}+\mathbf{B}'\right\|_F^2 = \operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right) + \operatorname{tr}\left(\mathbf{B}^2\right) \leq \operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right) + |\operatorname{tr}\left(\mathbf{B}^2\right)| \overset{(13)}{\leq} 2\operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right).$$

It remains to bound $\operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right)$ itself, which we compute directly:

$$\operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right) = \sum_{v=1}^{h+1}\sum_{l=1}^{h}\gamma_{v,l}^2\operatorname{tr}\left(\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{\Omega}_{v-l}^2\right)$$

for $\gamma_{v,l} = r_v'\mathbf{Q}_v^{-1}\alpha_{v,l}$, which is $O(T^{-1})$. On the other hand, iterated application of the trace-norm inequality shows that

$$|\operatorname{tr}\left(\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{\Omega}_{v-l}^2\right)| \leq C^2\operatorname{tr}\left(\mathbf{M}_v\right) = C^2(\bar{T}-K).$$

Hence $\operatorname{tr}\left(\mathbf{B}\mathbf{B}'\right) = O(T^{-2})\cdot O(T) = O(T^{-1})$. We conclude that the second term of 12 is also $O(T^{-1})$. □

Theorem 1. *Under assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors *and* Moderate-Dimensional Regime,

$$r'(\hat{\beta}-\beta) \xrightarrow{p} 0.$$

Proof. The result follows by combining Lemmas 6 and 7 □

Lemma 8 (Corollary A2.8 from Sølvsten (2020))**.** *Let $v_i = (v_{i1}', v_{i2}, v_{i3})'$ be a set of n iid vectors and $\Omega = Var\left(v_{11}\right)$. Let w_{ni} be a sequence of deterministic vectors of the same dimension as v_{i1}. Let M be a sequence of $n\times n$ symmetric matrices and $\bar{M} = M^2$ (the dependence of n is suppressed). Let*

$$W_n = \sum_i w_{ni}'v_{i1} \;+\; \sum_i\sum_{j\neq i} M_{ij}v_{i2}v_{j3}.$$

If the following conditions hold:

(i)

$$\sigma_n^2 = \sum_i w_{ni}'\Omega w_{ni} + \sum_i\sum_{j\neq i} M_{ij}^2\left(\mathbb{E}[v_{12}^2]\mathbb{E}[v_{13}^2] + \mathbb{E}[v_{12}v_{13}]^2\right) > c > 0;$$

(ii)

$$\mathbb{E}[v_{11}v_{11}'] = \Omega, \quad \sum_i w_{ni}'\Omega w_{ni} \le 1, \quad \max_i \|w_{ni}\|^2 \to 0;$$

(iii)

$$\mathbb{E}[v_{12}^4 + v_{13}^4] < \infty, \quad \sum_i \sum_{j\neq i} M_{ij}^2 \le 1, \quad \max_i \sum_{j\neq i} M_{ij}^2 \to 0, \quad \sum_i \sum_{j\neq i} \bar{M}_{ij}^2 \to 0;$$

(iv)

$$\mathbb{E}\big[\|v_{11}\|^2(v_{12}^2 + v_{13}^2)\big] < \infty;$$

then

$$\frac{W_n}{\sigma_n} \xrightarrow{d} \mathcal{N}(0,1).$$

Lemma 9 (Condition (iii) holds in the proportional regime)**.** *Let* $\Sigma = \mathrm{Var}(U) = \|\omega\|^2 + \frac{1}{2}\|B+B'\|_F^2$ *and* $\mathcal{B} := \tilde{B}/\sqrt{\Sigma}$, $\tilde{B} := \frac{1}{2}(B+B')$, *where* $\omega = r_v'\mathbf{Q}_v^{-1}\tilde{\mathbf{Z}}_{1,v}\mathbf{\Omega}_v$, $B = \Omega\sum_{v=1}^{h+1}\sum_{l=1}^{h}\gamma_{v,l}S_{v-l}'M_vS_v\Omega$ *and* $\gamma_{v,l} = r_v'Q_v^{-1}\alpha_{v,l}$ *are as defined in the proof of Lemma 6 . Suppose Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors *and* Moderate-Dimensional Regime *hold. Then*

$$\|\mathcal{B}\|^2, \max_i \sum_{j\neq i} \mathcal{B}_{ij}^2, \sum_i \sum_{j\neq i} (\mathcal{B}^2)_{ij}^2 = O(\bar{T}^{-1}).$$

Further, let $\bar{\mathcal{B}} = \sqrt{2}\tilde{B}/\left\|\tilde{B}\right\|_F$. *Then, if* $\|B+B'\|_F^2 = \Omega(\|\omega\|^2)$,

$$\|\bar{\mathcal{B}}\|^2, \max_i \sum_{j\neq i} \bar{\mathcal{B}}_{ij}^2, \sum_i \sum_{j\neq i} (\bar{\mathcal{B}}^2)_{ij}^2 = O(\bar{T}^{-1}).$$

Proof. Write $\tilde{B} = \frac{1}{2}(B+B')$ (symmetric). We show $\|\mathcal{B}\|_2^2 = \|\tilde{B}\|_2^2/\Sigma = O(\bar{T}^{-1})$; the three conditions then follow as in Steps 2–4 below. Throughout we use $\Sigma \ge \|\omega\|^2$ and set $\Pi_v := \|Q_v^{-1}r_v\|^2$.

Step 1: $\|\tilde{B}\|_2^2 \le C_1\sum_v \Pi_v$. By the triangle inequality and submultiplicativity of the spectral norm, and using that $\|M_v\|_2 = 1$, $\|S_v\|_2 = \|S_{v-l}'\|_2 = 1$, $\|\Omega\|_2^2 \le C$,

$$\|\tilde{B}\|_2 \le \|B\|_2 \le C\sum_{v=1}^{h+1}\sum_{l=1}^{h}|\gamma_{v,l}|.$$

Each $|\gamma_{v,l}| = |r_v'Q_v^{-1}\alpha_{v,l}| \le \|Q_v^{-1}r_v\|\,\|\alpha_{v,l}\| \le C\,\|Q_v^{-1}r_v\| = C\,\Pi_v^{1/2}$, using $\|\alpha_{v,l}\|^2 = \bar{T}\alpha_\ell'(\tilde{X}_v'\tilde{X}_v)^{-1}\alpha_\ell \le \|\alpha_\ell\|^2/c \le C^2/c$ (Lemma 1, Assumption *Bounded Eigenvalues*). Hence, by Cauchy–Schwarz:

$$\|\tilde{B}\|_2^2 \le C^2\Big(\sum_{v,l}|\gamma_{v,l}|\Big)^2 \le C^2h(h+1)\sum_{v,l}\gamma_{v,l}^2 \le C^2h^2(h+1)\,C^4/c\sum_v \Pi_v = O\left(\sum_v \Pi_v\right).$$

Step 2: denominator $\|\omega\|^2 \geq c\,\bar{T}\sum_v \Pi_v$.

$$\begin{aligned}\|\omega\|^2 =& \sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{\Omega}_v^2 \tilde{\mathbf{Z}}_{1,v} \mathbf{Q}_v^{-1} r_v \\ \geq& c \sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v}' \tilde{\mathbf{Z}}_{1,v} \mathbf{Q}_v^{-1} r_v \\ =& c\,\bar{T} \sum_v r_v' \mathbf{Q}_v^{-2} r_v \\ =& c\,\bar{T} \sum_v \Pi_v\end{aligned}$$

Step 3: the ratio. Combining Steps 1 and 2 with $\Sigma \geq \|\omega\|^2$, and noting the common factor $\sum_v \Pi_v > 0$ cancels,

$$\|\mathcal{B}\|_2^2 = \frac{\|\tilde{B}\|_2^2}{\Sigma} \leq \frac{\|\tilde{B}\|_2^2}{\|\omega\|^2} \leq \frac{O\left(\sum_v \Pi_v\right)}{\Omega(\bar{T}\sum_v \Pi_v)} = O(\bar{T}^{-1}). \tag{14}$$

Step 4: the two conditions. (Max row) For symmetric $\mathcal{B}$, $\max_i \sum_{j\neq i} \mathcal{B}_{ij}^2 \leq \max_i \|\mathcal{B}_{i\cdot}\|^2 \leq \|\mathcal{B}\|_2^2 = O(\bar{T}^{-1}) \to 0$ by (14). *($\mathcal{B}^2$)* With λ_k the eigenvalues of symmetric $\mathcal{B}$, $\sum_i \sum_{j\neq i} (\mathcal{B}^2)_{ij}^2 \leq \|\mathcal{B}^2\|_F^2 = \sum_k \lambda_k^4 \leq \|\mathcal{B}\|_2^2 \|\mathcal{B}\|_F^2 \leq O(\bar{T}^{-1}) \cdot 1 \to 0$. (For the last step, we used that $\|\mathcal{B}\|_F^2 \leq 1$, due to the normalization of dividing by $\sqrt{\Sigma}$.) The same statements follow for $\bar{\mathcal{B}}$ from the same reasoning under the stated assumption that $\|B + B'\|_F^2 = \Omega(\|\omega\|^2)$. □ □

Theorem 2. *Under assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, *and* No Influential Observations, *for some scalar sequence* Σ *defined in the proof of this theorem,*

$$\frac{r'(\hat{\beta} - \beta)}{\sqrt{\Sigma/(h+1)^2}} \xrightarrow{d} N(0,1).$$

Proof. Given Lemma 6, we need only prove

$$\frac{1}{\sqrt{\Sigma}} U := \frac{1}{\sqrt{\Sigma}} \sum_{v=1}^{h+1} r_v' \mathbf{Q}_v{}^{-1} \mathbf{Z}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon$$

to be asymptotically normal and the remainder term to be negligible. U can be re-written as follows:

$$\begin{aligned}U =& r_v' \sum_{v=1}^{h+1} \mathbf{Q}_v{}^{-1} \mathbf{Z}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon \\ =& r_v' \sum_{v=1}^{h+1} \mathbf{Q}_v{}^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon + r_v' \sum_{v=1}^{h+1} \mathbf{Q}_v{}^{-1} \sum_{l=1}^{h} \alpha_{v,l} \varepsilon' \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \varepsilon \\ =& r_v' \sum_{v=1}^{h+1} \mathbf{Q}_v{}^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{M}_v \mathbf{S}_v \varepsilon + \varepsilon' \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l} \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \varepsilon\end{aligned}$$

where $\gamma_{v,l} = r_v' \mathbf{Q}_v{}^{-1} \alpha_{v,l}$. Defining $\tilde{\varepsilon} = \mathbf{\Omega}^{-1}\varepsilon$,

$$\omega = \left(r_v' \sum_{v=1}^{h} \mathbf{Q}_v{}^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega} \right)'$$

and

$$\mathbf{B} = \mathbf{\Omega} \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l} \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega},$$

U can be written as:

$$U = \omega' \tilde{\varepsilon} + \tilde{\varepsilon}' \mathbf{B} \tilde{\varepsilon}$$

where conditionally on $\tilde{X}$, $\tilde{\varepsilon}$ is an iid vector with bounded fourth moment by assumption *Regular Errors*. We show asymptotic normality of the self-normalised sum by applying corollary A2.8 from Sølvsten (2020). This result applies to expressions like the one in equation A for a matrix B that is symmetric and has only zeros on the diagonal. The latter condition is satisfied by design in our case. The former condition can be easily met by rewriting

$$U = \omega' \tilde{\varepsilon} + \frac{1}{2} \tilde{\varepsilon}' (\mathbf{B} + \mathbf{B}') \tilde{\varepsilon}$$

The terms of this sum are uncorrelated conditionally on $\tilde{X}$:

$$\mathbb{E}\left[\omega_t \tilde{\varepsilon}_t (\mathbf{B}_{ts} + \mathbf{B}_{st}) \tilde{\varepsilon}_t \tilde{\varepsilon}_s\right] = \omega_t (\mathbf{B}_{ts} + \mathbf{B}_{st}) \mathbb{E}\left[\varepsilon_t^2 \varepsilon_s\right] = \mathbb{E}\left[\varepsilon_t^2\right] \mathbb{E}\left[\varepsilon_s\right] = 0$$

$$\mathbb{E}\left[(\mathbf{B}_{ts} + \mathbf{B}_{st}) \tilde{\varepsilon}_t \tilde{\varepsilon}_s (\mathbf{B}_{tu} + \mathbf{B}_{ut}) \tilde{\varepsilon}_t \tilde{\varepsilon}_u\right] = (\mathbf{B}_{ts} + \mathbf{B}_{st})(\mathbf{B}_{tu} + \mathbf{B}_{su}) \mathbb{E}\left[\tilde{\varepsilon}_t^2 \tilde{\varepsilon}_s \tilde{\varepsilon}_u\right] = 0$$

Hence

$$\begin{aligned}
\Sigma :=& Var\left(U\right) \\
=& \|\omega\|^2 + \sum_{t,s>t} (B_{ts} + B_{st})^2 \\
=& \|\omega\|^2 + \frac{1}{2} \|\mathbf{B} + \mathbf{B}'\|_F^2
\end{aligned}$$

Now we apply corollary A2.8 from Sølvsten (2020) to show that

$$\frac{U}{\sqrt{\Sigma}} = \frac{1}{\sqrt{\Sigma}} \left(\omega' \tilde{\varepsilon} + \frac{1}{2} \tilde{\varepsilon}' \left(\mathbf{B} + \mathbf{B}' \right) \tilde{\varepsilon} \right) \xrightarrow{d} N(0,1)$$

The weights w_{nt} and M_{st} in their notation correspond to $\omega_t / \sqrt{\Sigma}$ and $\frac{1}{2}(B_{st} + B_{ts}) / \sqrt{\Sigma}$ in my notation, respectively.

1. condition (i) translates to $\frac{1}{\Sigma}\left(\|\omega\|^2 + \frac{1}{2} \|\mathbf{B} + \mathbf{B}'\|_F^2 \right) > c > 0$ which is automatically satisfied here because $\frac{1}{\Sigma}\left(\|\omega\|^2 + \frac{1}{2} \|\mathbf{B} + \mathbf{B}'\|_F^2 \right) = 1$

2. The first part of condition (ii) here translates to $\frac{1}{\Sigma}\|\omega\|^2 \leq 1$ which is also automatically satisfied. The second part of condition (ii) translates to $\max_t \omega_t^2/\Sigma \to 0$. First note that the first h elements of ω are already equal to zero.

   Since $\mathbf{M}_v\tilde{\mathbf{Z}}_{1,v} = \tilde{\mathbf{Z}}_{1,v}$ (Lemma 1, already used to establish (1) in the proof of Lemma 3), the t-th column of $\tilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v$ equals $\tilde{\mathbf{Z}}'_{1,v}$'s t-th column, namely $\tilde{z}_{1,t}$, so that the t-th element of ω (for t in subsample $v = u(t)$) takes the simple form

$$\omega_t = r'_v \mathbf{Q}_v^{-1} \tilde{z}_{1,t}\, \sigma_t.$$

   We bound $\max_t \omega_t^2/\Sigma$ using three facts.

   (a) By the Cauchy-Schwarz inequality, $\lambda_{\min}(\mathbf{Q}_v) \geq \bar{T}$ (Lemma 4), and $\sigma_t^2 \leq C$ (Assumption *Regular Errors*),

$$\omega_t^2 \leq \sigma_t^2 \left\|\mathbf{Q}_v^{-1} r_v\right\|_2^2 \|\tilde{z}_{1,t}\|_2^2 \leq C\bar{T}^{-2}\|r_v\|^2\|\tilde{z}_{1,t}\|^2.$$

   (b) By equation (17) in the proof of Lemma 10, $\|\tilde{z}_{1,t}\|^2 = f'_t \left(\mathbf{A}(\tilde{\mathbf{X}}'_{u(t)}\tilde{\mathbf{X}}_{u(t)}/\bar{T})^{-1}\mathbf{A}'\right)^{-1} f_t \leq \frac{C}{c}\|f_t\|^2$, using $\lambda_{\min}(\mathbf{A}\mathbf{A}') \geq c$ (Assumption *No Influential Observations*) and Assumption *Bounded Eigenvalues*.

   (c) Since $\mathbf{M}_v\tilde{\mathbf{Z}}_{1,v} = \tilde{\mathbf{Z}}_{1,v}$ and $\mathbf{\Omega}_v^2 \succeq c\,\mathbf{I}_{\bar{T}}$ (Assumption *Regular Errors*), $\tilde{\mathbf{Z}}'_{1,v}\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\tilde{\mathbf{Z}}_{1,v} = \tilde{\mathbf{Z}}'_{1,v}\mathbf{\Omega}_v^2\tilde{\mathbf{Z}}_{1,v} \succeq c\,\tilde{\mathbf{Z}}'_{1,v}\tilde{\mathbf{Z}}_{1,v} = c\bar{T}\mathbf{I}_{h+1}$ (Lemma 1). Together with $\lambda_{\max}(\mathbf{Q}_v) \leq C\bar{T}$ (Lemma 4), which gives $\mathbf{Q}_v^{-2} \succeq (C\bar{T})^{-2}\mathbf{I}_{h+1}$, this yields, for every v,

$$\Sigma \geq \|\omega\|^2 = \sum_w r'_w \mathbf{Q}_v^{-1}\tilde{\mathbf{Z}}'_{1,w}\mathbf{\Omega}_w^2\tilde{\mathbf{Z}}_{1,w}\mathbf{Q}_v^{-1} r_w \geq c\bar{T}\sum_w r'_w\mathbf{Q}_w^{-2}r_w \geq c\bar{T}\, r'_v\mathbf{Q}_v^{-2}r_v \geq \frac{c}{C^2\bar{T}}\|r_v\|^2,$$

   using that every summand in $\sum_w$ is non-negative.

Combining (a)-(c): whenever $r_v \neq 0$, for t in subsample v,

$$\frac{\omega_t^2}{\Sigma} \leq \frac{C\bar{T}^{-2}\|r_v\|^2 \cdot \frac{C}{c}\|f_t\|^2}{\frac{c}{C^2\bar{T}}\|r_v\|^2} = C'\,\frac{\|f_t\|^2}{\bar{T}}$$

for a constant C' not depending on t, v or T (and, if $r_v = 0$, $\omega_t = 0$ by (a) so the same bound holds trivially). Hence

$$\max_t \frac{\omega_t^2}{\Sigma} \leq C'\,\frac{\max_t\|f_t\|^2}{\bar{T}} \to 0,$$

which vanishes by Assumption *No Influential Observations*. (This replaces the derivation via the Frisch–Waugh–Lovell theorem and $\tilde{r}_v$ used in an earlier version of this proof, which controlled only the influence of observations in the r-direction; the argument above controls the full cross-leverage f_t of Definition 1, which is what condition (ii) actually requires once $h \geq 1$.)

3. The third condition translates to $\mathbb{E}[\varepsilon_t^4] < \infty$ (satisfied by assumption *Regular Errors*), $\frac{1}{\Sigma}\left\|\frac{1}{2}(\mathbf{B}+\mathbf{B}')\right\|_F^2 \leq 1$ (satisfied by construction), $\max_i \frac{1}{\Sigma}\sum_j \left(\frac{\mathbf{B}_{ij}+\mathbf{B}_{ji}}{2}\right)^2 \to 0$ and, letting $\bar{\mathbf{B}} = \frac{1}{4}(\mathbf{B}+\mathbf{B}')^2$, $\frac{1}{\Sigma^2}\sum_i \sum_{j\neq 1} \bar{\mathbf{B}}_{ij}^2 \to 0$. The latter two conditions are proven to hold by Lemma 9. (Note that Lemma 9 operates under the more general assumption that $\limsup \frac{K}{T} < 1$.)

4. The fourth condition amounts to bounded fourth moments on the ε_t, which is assumed in Assumption *Regular Errors*.

Thus, the conditions of Lemma 8 are satisfied, proving that $\frac{1}{\sqrt{\Sigma}}U \xrightarrow{d} N(0,1)$.

To show that the remainder

$$\frac{1}{\sqrt{\Sigma}}\sum_{v=1}^{h+1} r_v'\left(\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} - \mathbf{Q}_v{}^{-1}\right)\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon = \frac{U}{\sqrt{\Sigma}}\frac{\sum_{v=1}^{h+1} r_v'\left(\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} - \mathbf{Q}_v{}^{-1}\right)\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon}{U}$$

is $o_p(1)$, it suffices to show that

$$\frac{\sum_{v=1}^{h+1} r_v'\left(\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} - \mathbf{Q}_v{}^{-1}\right)\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon}{U} = \frac{\sum_{v=1}^{h+1} r_v'\left(\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} - \mathbf{Q}_v{}^{-1}\right)\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon}{\sum_{v=1}^{h+1} r_v'\mathbf{Q}_v{}^{-1}\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{S}_v\varepsilon}$$

is $o_p(1)$. This is the case as we know from the proof of Lemma 5 that

$$\lambda_{\max}\left(\left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} - \mathbf{Q}_v{}^{-1}\right) = O_p(T^{-3/2}),$$

whereas from Lemma 4 we know that

$$\lambda_{\min}\left(\mathbf{Q}_v^{-1}\right) = \lambda_{\max}\left(\mathbf{Q}_v\right)^{-1} = \Omega(T^{-1}).$$

□

Lemma 10 (Bounded average fourth moment of $z_{1,r(v,t)}$)**.** *Under assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, No Influential Observations *and* Bounded Cross-Leverage *, for every* $v \in \{1,\ldots,h+1\}$,

$$\frac{1}{\bar{T}}\sum_{t=1}^{\bar{T}} \mathbb{E}\left[\left\|z_{1,r(v,t)}\right\|^4\right] = O(1),$$

$$\frac{1}{\bar{T}}\sum_{t=1}^{\bar{T}} \mathbb{E}\left[\left\|\hat{z}_{1,r(v,t)}\right\|^4\right] = O(1).$$

Proof. Fix a subsample $v \in \{1, \dots, h+1\}$. Row t of $M_v Z_{1,v}$ admits the decomposition

$$\hat{z}_{1,r(v,t)} = \tilde{z}_{1,r(v,t)} + \sum_{\ell=1}^{h} \psi_{v\ell t}\, \alpha_{v,\ell}, \qquad \psi_{v\ell t} := M_{v,t\cdot} S_{v-\ell}\varepsilon, \tag{15}$$

where we used that $M_v \tilde{Z}_{1,v} = \tilde{Z}_{1,v}$.

By the c_r-inequality $\|a+b\|^4 \le 8(\|a\|^4 + \|b\|^4)$ applied to (15), and then again to the sum over ℓ,

$$\|\hat{z}_{1,r(v,t)}\|^4 \le 8\,\|\tilde{z}_{1,r(v,t)}\|^4 + 8\,h^3 \sum_{\ell=1}^{h} |\psi_{v\ell t}|^4\, \|\alpha_{v,\ell}\|^4. \tag{16}$$

We bound the average over t of each term separately.

Bound on the non-random part. Write $\Sigma_v := (\tilde{X}_v'\tilde{X}_v/\bar{T})^{1/2}$. Let Q_v be the matrix from the proof of Lemma 1, i.e. the columns of Q_v form an orthonormal basis for $\mathrm{col}(\Sigma_v^{-1}A')$, and the orthogonal projection onto this subspace is

$$Q_v Q_v' = \Sigma_v^{-1}A'(A\Sigma_v^{-2}A')^{-1}A\Sigma_v^{-1} = (\tilde{X}_v'\tilde{X}_v/\bar{T})^{-1/2}A'\big(A(\tilde{X}_v'\tilde{X}_v/\bar{T})^{-1}A'\big)^{-1}A(\tilde{X}_v'\tilde{X}_v/\bar{T})^{-1/2}.$$

Let $f_{r(v,t)}$ be as defined in Assumption *Bounded Cross-Leverage*. Since $\Theta_{v,1} = \Sigma_v^{-1}Q_v$,

$$\begin{aligned}\|\tilde{z}_{1,r(v,t)}\|^2 &= \tilde{x}_{r(v,t)}'\Theta_{v,1}\Theta_{v,1}'\tilde{x}_{r(v,t)} = \tilde{x}_{r(v,t)}'\Sigma_v^{-1}Q_vQ_v'\Sigma_v^{-1}\tilde{x}_{r(v,t)} \\ &= f_{r(v,t)}'\big(A(\tilde{X}_v'\tilde{X}_v/\bar{T})^{-1}A'\big)^{-1}f_{r(v,t)} \;\le\; \frac{\|f_{r(v,t)}\|^2}{\lambda_{\min}\big(A(\tilde{X}_v'\tilde{X}_v/\bar{T})^{-1}A'\big)}.\end{aligned} \tag{17}$$

For the denominator, for any unit vector $u \in \mathbb{R}^{h+1}$,

$$\lambda_{\min}\left(A(\tilde{X}_v'\tilde{X}_v/\bar{T})^{-1}A'\right) \ge \frac{c}{C},$$

using $\lambda_{\min}\left(\left(\tilde{X}_v'\tilde{X}_v/\bar{T}\right)^{-1}\right) = \lambda_{\max}\left(\left(\tilde{X}_v'\tilde{X}_v/\bar{T}\right)\right)^{-1} \ge 1/C$ and $\lambda_{\min}(AA') \ge c > 0$. Thus,

$$\|\tilde{z}_{1,r(v,t)}\|^4 \;\le\; \frac{C^2}{c^2}\,\|f_{r(v,t)}\|^4. \tag{18}$$

Averaging over t and applying Assumption *Bounded Cross-Leverage*,

$$\frac{1}{\bar{T}}\sum_{t=1}^{\bar{T}} \|\tilde{z}_{1,r(v,t)}\|^4 \;\le\; \frac{C^2}{c^2}\cdot\frac{1}{\bar{T}}\sum_{t=1}^{\bar{T}}\|f_{r(v,t)}\|^4 \;=\; O(1). \tag{19}$$

Feedback part. Fix $\ell \in \{1, \dots, h\}$. By definition

$$\psi_{v\ell t} = M_{v,t\cdot}S_{v-\ell}\varepsilon = \sum_{s=1}^{\bar{T}} M_{v,ts}\,\varepsilon_{v-\ell+(s-1)(h+1)},$$

a linear combination of independent, mean-zero random variables which have variances in $[c, C]$ and uniformly bounded fourth moments $E[\varepsilon_t^4] \le C^2$ (Assumption **??**). Rosenthal's inequality therefore gives, for a universal constant C_R,

$$\begin{aligned} E\big[\psi_{v\ell t}^4\big] &\le C_R\Big(\sum_s M_{v,ts}^4\, E[(S_{v-\ell}\varepsilon)_s^4] + \Big(\sum_s M_{v,ts}^2\, E[(S_{v-\ell}\varepsilon)_s^2]\Big)^2\Big) \\ &\le C_R\big(C^2 + C^2\big)\Big(\sum_s M_{v,ts}^2\Big)^2, \end{aligned} \tag{20}$$

where the last step uses $M_{v,ts}^4 \le M_{v,ts}^2$ (valid since $|M_{v,ts}| \le 1$ for a projection matrix) to merge the two sums. Because M_v is an orthogonal projection,

$$\sum_s M_{v,ts}^2 = \|M_{v,\cdot t}\|^2 = M_{v,tt} \le \|M_v\|_2^2 = 1,$$

so (20) yields the uniform bound

$$E\big[\psi_{v\ell t}^4\big] \le 2C_R C^2 =: C' = O(1), \qquad \text{uniformly in } v, \ell, t. \tag{21}$$

Recalling from Lemma 1 that $\|\alpha_{v,\ell}\|^2 = \bar{T}\,\alpha_\ell'(\tilde{X}_v'\tilde{X}_v)^{-1}\alpha_\ell \le \|\alpha_\ell\|^2/c \le C^2/c$, the feedback term of (16) obeys

$$\frac{8h^3}{\bar{T}}\sum_{t=1}^{\bar{T}}\sum_{\ell=1}^{h} E[|\psi_{v\ell t}|^4]\,\|\alpha_{v,\ell}\|^4 \le 8h^3 \cdot h \cdot C' \cdot \frac{C^4}{c^2} = \frac{8h^4 C' C^4}{c^2} = O(1). \tag{22}$$

Likewise,

$$z_{1,r(v,t)} = \tilde{z}_{1,r(v,t)} + \sum_{\ell=1}^{h} \varepsilon_{r(v,t)-\ell}\,\alpha_{v,\ell}, \tag{23}$$

so the c_r-inequality, exactly as in (16), gives

$$\|z_{1,r(v,t)}\|^4 \le 8\,\|\tilde{z}_{1,r(v,t)}\|^4 + 8\,h^3 \sum_{\ell=1}^{h} |\varepsilon_{r(v,t)-\ell}|^4\,\|\alpha_{v,\ell}\|^4. \tag{24}$$

The non-random part is identical to (19), and is hence $O(1)$ on average. For the feedback part, $E[|\varepsilon_{r(v,t)-\ell}|^4] \le C^2$ replaces the Rosenthal bound (21), so the same computation as in (22) yields

$$\frac{8h^3}{\bar{T}}\sum_{t=1}^{\bar{T}}\sum_{\ell=1}^{h} E[|\varepsilon_{r(v,t)-\ell}|^4]\,\|\alpha_{v,\ell}\|^4 \le 8h^3 \cdot h \cdot C^2 \cdot \frac{C^4}{c^2} = \frac{8h^4 C^6}{c^2} = O(1). \tag{25}$$

Taking expectations in (24) and averaging therefore gives $\frac{1}{\bar{T}}\sum_{t=1}^{\bar{T}} E\|z_{1,r(v,t)}\|^4 = O(1)$ uniformly in v as well.

□

Lemma 11 (Any sandwich with consistent meat is consistent for Σ)**.** *Define the population meat of subsample v*

$$\mathbf{\Psi}_v := \mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{Z}_{1,v}$$

and let $\hat{\mathbf{\Psi}}_1, \ldots, \hat{\mathbf{\Psi}}_{h+1}$ be any (possibly random) $(h+1)\times(h+1)$ matrices satisfying

$$\frac{1}{\bar{T}}\left\|\hat{\mathbf{\Psi}}_v - \mathbf{\Psi}_v\right\|_2 \xrightarrow{p} 0, \qquad v = 1, \ldots, h+1. \tag{26}$$

Then, under Assumptions Feedback Decomposition*,* Bounded Eigenvalues*,* Regular Errors *and* Moderate-Dimensional Regime*,*

$$\frac{\sum_{v=1}^{h+1} r'_v \left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1}\hat{\mathbf{\Psi}}_v\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} r_v}{\Sigma} \xrightarrow{p} 1.$$

Proof. Step 1: replacing $\hat{\mathbf{\Psi}}_v$ by $\mathbf{\Psi}_v$ is harmless. By Lemma 3 and Assumption *Regular Errors*,

$$\begin{aligned}\lambda_{\min}\left(\frac{1}{\bar{T}}\mathbf{\Psi}_v\right) = \lambda_{\min}\left(\frac{1}{\bar{T}}\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{Z}_{1,v}\right) &\geq \lambda_{\min}\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T}\right)\lambda_{\min}\left(\mathbf{\Omega}_v^2\right)\\ &\geq (1-o_p(1))\,c.\end{aligned}$$

Writing $u_v := \left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} r_v$, it follows that for every v,

$$\frac{\left|u'_v\left(\hat{\mathbf{\Psi}}_v - \mathbf{\Psi}_v\right)u_v\right|}{u'_v\mathbf{\Psi}_v u_v} \leq \frac{\left\|\hat{\mathbf{\Psi}}_v - \mathbf{\Psi}_v\right\|_2\|u_v\|^2}{\lambda_{\min}\left(\mathbf{\Psi}_v\right)\|u_v\|^2} = \frac{o_p(\bar{T})}{c\,\bar{T}(1-o_p(1))} = o_p(1),$$

where the numerator uses (26). As the $h+1$ summands $u'_v\mathbf{\Psi}_v u_v$ are non-negative, summing over v preserves the bound, so

$$\sum_v r'_v\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1}\hat{\mathbf{\Psi}}_v\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} r_v = (1+o_p(1))\sum_v r'_v\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1}\mathbf{\Psi}_v\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} r_v. \tag{27}$$

It therefore suffices to show that the right-hand side of (27) converges in ratio to Σ.

Step 2: replacing $\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}$ by its expectation. By the same argument, we may replace either or both of the $\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}$ by their expectation $\mathbf{Q}_v$: this is because $\lambda_{\max}\left((\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v})^{-1} - \mathbf{Q}_v^{-1}\right) = O_p(\bar{T}^{-3/2})$ (Lemma 5) whereas $\lambda_{\min}\left(\mathbf{Q}_v^{-1}\right) = \lambda_{\max}\left(\mathbf{Q}_v\right)^{-1} = \Omega(\bar{T}^{-1})$ (Lemma 4).

Step 3: identifying the limit. Using $\mathbf{M}_v\tilde{\mathbf{Z}}_{1,v} = \tilde{\mathbf{Z}}_{1,v}$, which holds because the transformation $\mathbf{\Theta}_v$ orthogonalises the regressors, we decompose

$$\begin{aligned}\sum_v r'_v\mathbf{Q}_v^{-1}\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{Z}_{1,v}\mathbf{Q}_v^{-1}r_v = &\sum_v r'_v\mathbf{Q}_v^{-1}\tilde{\mathbf{Z}}'_{1,v}\mathbf{\Omega}_v^2\tilde{\mathbf{Z}}_{1,v}\mathbf{Q}_v^{-1}r_v +\\ &\sum_v\sum_l\sum_m r'_v\mathbf{Q}_v^{-1}\alpha_{v,l}\varepsilon'\mathbf{S}'_{v-l}\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{S}_{v-m}\varepsilon\alpha'_{v,m}\mathbf{Q}_v^{-1}r_v +\\ &\sum_v\sum_l r'_v\mathbf{Q}_v^{-1}\tilde{\mathbf{Z}}'_{1,v}\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{S}'_{v-l}\varepsilon\alpha'_{v,l}\mathbf{Q}_v^{-1}r_v.\end{aligned}$$

The first term is exactly the linear part of Σ:

$$\sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{\Omega}_v^2 \tilde{\mathbf{Z}}_{1,v} \mathbf{Q}_v^{-1} r_v = \|\omega\|^2,$$

and, for future reference, we note the lower bound

$$\begin{aligned}
\|\omega\|^2 =& \sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{\Omega}_v^2 \tilde{\mathbf{Z}}_{1,v} \mathbf{Q}_v^{-1} r_v \\
\geq& \sum_v \|r_v\|^2 \lambda_{\max}\left(\mathbf{Q}_v\right)^{-2} \lambda_{\min}\left(\tilde{\mathbf{Z}}_{1,v}' \tilde{\mathbf{Z}}_{1,v}\right) \lambda_{\min}\left(\mathbf{\Omega}_v^2\right) \\
\geq& c\bar{T} \sum_v \|r_v\|^2 \Omega(\bar{T}^{-2}) \\
=& \sum_v \|r_v\|^2 \Omega(\bar{T}^{-1}).
\end{aligned} \tag{28}$$

We turn to the second term,

$$\sum_v \sum_l \sum_m r_v' \mathbf{Q}_v^{-1} \alpha_{v,l} \varepsilon' \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{\Omega}_v{}^2 \mathbf{M}_v \mathbf{S}_{v-m} \varepsilon \alpha_{v,m}' \mathbf{Q}_v^{-1} r_v, \tag{29}$$

and show that it approximates the quadratic part of Σ, namely $\frac{1}{2}\|\mathbf{B}+\mathbf{B}'\|_F^2 = \operatorname{tr}(\mathbf{B}\mathbf{B}') + \operatorname{tr}(\mathbf{B}^2)$. Focusing first on the terms with $m = l$, and recalling $\gamma_{v,l} = r_v' \mathbf{Q}_v^{-1} \alpha_{v,l}$,

$$\begin{aligned}
\mathbb{E}\left[\sum_{v,l} \gamma_{v,l}^2 \varepsilon' \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l} \varepsilon\right] &= \sum_{v,l} \gamma_{v,l}^2 \operatorname{tr}\left(\mathbf{\Omega} \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l} \mathbf{\Omega}\right) \\
&= \operatorname{tr}(\mathbf{B}\mathbf{B}').
\end{aligned}$$

To explain the last step:

$$\begin{aligned}
\mathbf{B}\mathbf{B}' =& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \sum_{q=1}^{h+1} \sum_{m=1}^{h} \gamma_{v,l} \gamma_{q,m} \mathbf{\Omega} \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_q' \mathbf{M}_q \mathbf{S}_{q-l} \mathbf{\Omega} \\
=& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \sum_{m=1}^{h} \gamma_{v,l} \gamma_{v,m} \mathbf{\Omega} \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-m} \mathbf{\Omega}
\end{aligned}$$

Using linearity and cyclicity of the trace:

$$\begin{aligned}
\operatorname{tr}(\mathbf{B}\mathbf{B}') =& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \sum_{m=1}^{h} \gamma_{v,l} \gamma_{v,m} \operatorname{tr}\left(\mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-m} \mathbf{\Omega}^2 \mathbf{S}_{v-l}'\right) \\
=& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l}^2 \operatorname{tr}\left(\mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l} \mathbf{\Omega}^2 \mathbf{S}_{v-l}'\right)
\end{aligned}$$

The second equality uses the fact that for $l, m \in \{1, \ldots, h\}$ such that $l \neq m$: $\mathbf{S}_{v-m} \mathbf{\Omega}^2 \mathbf{S}_{v-l}' = \mathbf{0}_{\bar{T}\times\bar{T}}$ for all $v \in \{1, \ldots, h+1\}$.

Still focusing on the terms of (29) with $m = l$, we derive the order of magnitude of its deviation from the expectation:

$$\varepsilon' \sum_{v,l} \gamma_{v,l}^2 \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l} \varepsilon - \mathbb{E}\left[\sum_{v,l} \gamma_{v,l}^2 \varepsilon' \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l} \varepsilon \right]. \tag{30}$$

Following Lemma 2, each term of this sum is $O_p(\left\|\gamma_{v,l}^2 \mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l}\right\|_F)$. Using the structure of $\mathbf{S}_v$ and a property of the Frobenius norm,

$$\left\|\mathbf{S}_{v-l}' \mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l}\right\|_F \leq \|\mathbf{M}_v\|_2 \left\|\mathbf{\Omega}_v^2\right\|_2 \|\mathbf{M}_v\|_F = O(\sqrt{\bar{T}}).$$

In other words, (30) is $O_p(\sqrt{\bar{T}} \sum_{v,l} \gamma_{v,l}^2)$. To show that this devition from the mean is of a lower order than the mean itself, we obtain a lower bound for $\operatorname{tr}(\mathbf{B}\mathbf{B}')$ as follows:

$$\begin{aligned}
\operatorname{tr}(\mathbf{B}\mathbf{B}') =& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l}^2 \operatorname{tr}\left(\mathbf{M}_v \mathbf{S}_v \mathbf{\Omega}^2 \mathbf{S}_v' \mathbf{M}_v \mathbf{S}_{v-l} \mathbf{\Omega}^2 \mathbf{S}_{v-l}'\right) \\
=& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l}^2 \operatorname{tr}\left(\mathbf{M}_v \mathbf{\Omega}_v^2 \mathbf{M}_v \mathbf{\Omega}_{v-l}^2\right) \\
\geq& \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l}^2 c^2 \operatorname{tr}(\mathbf{M}_v) \\
=& (\bar{T} - K) c^2 \sum_{v=1}^{h+1} \sum_{l=1}^{h} \gamma_{v,l}^2
\end{aligned} \tag{31}$$

This shows that (30) is $o_p(\operatorname{tr}(\mathbf{B}\mathbf{B}'))$. We turn to the summands of (29) with $m \neq l$, to which the same reasoning applies, except that their expectation is zero. Thus, by Lemma 2, these terms are $O_p(\gamma_{v,l}\gamma_{v,m}\sqrt{\bar{T}})$ each and hence $o_p(\operatorname{tr}(\mathbf{B}\mathbf{B}'))$.

Finally, we are left with the third set of terms, which we bound as follows:

$$\begin{aligned}
\sum_v r_v' \mathbf{Q}_v^{-1} \tilde{\mathbf{Z}}_{1,v}' \mathbf{\Omega}_v^2 \sum_l \mathbf{S}_{v-l} \varepsilon \alpha_{v,l}' \mathbf{Q}_v^{-1} r_v \leq& \sum_v \|r_v\|^2 \lambda_{\min}(\mathbf{Q}_v)^{-2} \sqrt{\bar{T}} \left\| \tilde{\mathbf{Z}}_{1,v}' \mathbf{\Omega}_v \sum_l \mathbf{S}_{v-l} \varepsilon \alpha_{v,l}' / \sqrt{\bar{T}} \right\|_2 \\
=& \sum_v \|r_v\|^2 O(\bar{T}^{-2}) \bar{T}^{1/2} O_p(1),
\end{aligned}$$

where we used Lemmas 2 and 4. Remembering from (28) that $\|\omega\|^2 \geq \sum_v \|r_v\|^2 \Omega(\bar{T}^{-1})$, this last term is negligible relative to $\|\omega\|^2$ as well.

Collecting the three terms, the right-hand side of (27) equals $\|\omega\|^2 + \frac{1}{2}\|\mathbf{B} + \mathbf{B}'\|_F^2 + o_p(\Sigma) = \Sigma(1 + o_p(1))$, which together with (27) proves the claim. □

Lemma 12. *Let $\{X_i\}_{i=1}^n$ be random variables satisfying*

$$\sup_i E\left[|X_i|^{2+\delta}\right] \leq M < \infty \quad \textit{for some } \delta > 0.$$

Then $\max_i |X_i| = o_p(n^{1/2})$.

Proof. Fix $\varepsilon > 0$. By the union bound,

$$P\left(\max_{1\le i\le n} |X_i| > \varepsilon n^{1/2}\right) \le \sum_{i=1}^{n} P\big(|X_i| > \varepsilon n^{1/2}\big).$$

Apply Markov's inequality to $|X_i|^{2+\delta}$:

$$P\big(|X_i| > \varepsilon n^{1/2}\big) = P\big(|X_i|^{2+\delta} > \varepsilon^{2+\delta} n^{(2+\delta)/2}\big) \le \frac{E\big[|X_i|^{2+\delta}\big]}{\varepsilon^{2+\delta}\, n^{(2+\delta)/2}} \le \frac{M}{\varepsilon^{2+\delta}\, n^{1+\delta/2}}.$$

Summing over $i = 1, \dots, n$,

$$P\left(\max_{1\le i\le n} |X_i| > \varepsilon n^{1/2}\right) \le n \cdot \frac{M}{\varepsilon^{2+\delta}\, n^{1+\delta/2}} = \frac{M}{\varepsilon^{2+\delta}}\, n^{-\delta/2}.$$

Since $\delta > 0$, the right-hand side tends to 0 as $n \to \infty$. As $\varepsilon > 0$ was arbitrary, $\max_i |X_i| = o_p(n^{1/2})$. □

Lemma 13 (Maximum row norm of $\mathbf{M}_v\mathbf{Z}_{1,v}$)**.** *Under assumptions* Feedback Decomposition, Moderate-Dimensional Regime, Bounded Eigenvalues, Regular Errors, No Influential Observations *and* Bounded Cross-Leverage,

$$\max_{v\in\{1,\dots,h+1\}} \max_{t\in\{1,\dots,\bar{T}\}} \|\tilde{z}_{1,r(v,t)}\| = o_p(\bar{T}^{1/2})$$

and

$$\max_{v\in\{1,\dots,h+1\}} \max_{t\in\{1,\dots,\bar{T}\}} \|\hat{z}_{1,r(v,t)}\| = o_p(\bar{T}^{1/2}).$$

Proof. Fix v and recall that

$$\hat{z}_{1,r(v,t)} = \widetilde{\mathbf{Z}}'_{1,v} M_{v,\cdot t} + \sum_{\ell=1}^{h} \psi_{v\ell t}\, \alpha_{v,\ell}, \qquad \psi_{v\ell t} := M_{v,t\cdot} S_{v-\ell}\varepsilon. \tag{32}$$

By the triangle inequality and $\|\alpha_{v,\ell}\| \le C$,

$$\max_t \|\hat{z}_{1,r(v,t)}\| \le \max_t \|\widetilde{\mathbf{Z}}'_{1,v} M_{v,\cdot t}\| + C \sum_{\ell=1}^{h} \max_t |\psi_{v\ell t}|. \tag{33}$$

Bound on the non-random part. From equation 17 in Lemma(10), $\|\widetilde{\mathbf{Z}}'_{1,v} M_{v,\cdot t}\|^2 \le (C^2/c^2)\|f_{r(v,t)}\|^2$ because $\lambda_{\min}(AA') > c$. Because $\frac{1}{\bar{T}}\sum_t \|f_{r(v,t)}\|^4 < C$, by Assumption *Bounded Cross-Leverage*, $\max_t \|f_{r(v,t)}\|^4 \le \sum_t \|f_{r(v,t)}\|^4 < C\bar{T}$. Thus $\max_t \|f_{r(v,t)}\|^2 = O(\bar{T}^{\frac{1}{2}}) = o(\bar{T})$. This establishes that

$$\max_t \|\tilde{z}_{1,r(v,t)}\|^2 = \max_t \|\widetilde{\mathbf{Z}}'_{1,v} M_{v,\cdot t}\|^2 = o(\bar{T}).$$

Bound on the feedback part. Fix ℓ. Since $\psi_{v\ell t} = \sum_s M_{v,ts}(S_{v-\ell}\varepsilon)_s$ is a linear combination of independent mean-zero random variables with variances in $[c, C]$ and bounded $2+\delta$

moment (for some $\delta > 0$), Rosenthal's inequality is applicable and shows that $\psi_{v\ell t}$ also has moments greater than 2 :

$$\begin{aligned}\mathbb{E}[|\psi_{v\ell t}|^{2+\delta}] \leq & C\left(\sum_s M_{v,ts}^{2+\delta}\mathbb{E}\left[|\varepsilon_{v-\ell+(s-1)(h+1)}|^{2+\delta}\right] + \sum_s M_{v,ts}^2\left(\mathbb{E}\left[\varepsilon_{v-\ell+(s-1)(h+1)}^2\right]\right)^{1+\delta/2}\right)\\ \leq & C(C^2 + C^{1+\delta/2})\sum_s M_{v,ts}^2 = C(C^2 + C^{1+\delta/2})\|M_{v,\cdot t}\|^2\\ \leq & C(C^2 + C^{1+\delta/2}),\end{aligned}$$

The last inequality uses $\|M_{v,\cdot t}\|^2 \leq \|M_v\|_2^2 = 1$. Lemma **??** then gives that $\max \psi_{vlt} = o_p(\bar{T}^{1/2})$. □

Lemma 14. *Let* $h_t = 1 - x_t'(X_{u(t)}'X_{u(t)})^{-1}x_t$. *Under assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors *and* Moderate-Dimensional Regime, No Influential Observations, Bounded Cross-Leverage *and* Leave-One-Out Conditioning,

$$\max_{1\leq t\leq T}\frac{1}{h_t} = O_p(1)$$

Proof. We show first that the statement holds for $\tilde{h}_t := \tilde{\mathbf{H}}_{u(t),tt}$ where

$$\tilde{\mathbf{H}} = \mathbf{I}_{\bar{T}} - \tilde{\mathbf{X}}_{u(t)}(\tilde{\mathbf{X}}_{u(t)}'\tilde{\mathbf{X}}_{u(t)})^{-1}\tilde{\mathbf{X}}_{u(t)}'.$$

For each t, let $\mathbf{G}_{u(t),-t} = \frac{1}{T}\tilde{\mathbf{X}}_{u(t),-t}'\tilde{\mathbf{X}}_{u(t),-t}$, where $\tilde{\mathbf{X}}_{u(t),-t}$ is as defined in assumption *Leave-One-Out Conditioning*. Applying the Sherman-Morrison formula to the rank-one decomposition $\frac{1}{T}\tilde{\mathbf{X}}_{u(t)}'\tilde{\mathbf{X}}_{u(t)} = \mathbf{G}_{u(t),-t} + \frac{1}{T}\tilde{x}_t\tilde{x}_t'$ gives

$$\tilde{h}_t = \frac{1}{1+\tau_t}, \qquad \tau_t := \frac{1}{T}\,\tilde{x}_t'\mathbf{G}_{u(t),-t}^{-1}\tilde{x}_t \;\geq\; 0,$$

Using $\lambda_{\min}(\mathbf{G}_{u(t),-t}) > c$ and $\|\tilde{x}_t\|^2 \leq \left\|\tilde{\mathbf{X}}_{u(t)}\right\|_2^2 = \lambda_{\max}(\tilde{\mathbf{X}}_{u(t)}'\tilde{\mathbf{X}}_{u(t)}) < TC$, $\max_t \tau_t \;\leq\; \frac{C}{c}$, which establishes that $\min_t \tilde{h}_t$ is bounded away from zero. Next we, let $m_t = \mathbf{M}_{v,tt}$. As $\mathrm{col}(\mathbf{Z}_{2,v}) \subseteq \mathrm{col}(\mathbf{X}_v)$, $\mathbf{M}_v - \tilde{\mathbf{H}}_v$ is positive semi-definite implying in particular that $m_t \geq \tilde{h}_t$. Lastly, as $\mathrm{col}(\mathbf{Z}_{2,v}) \subseteq \mathbf{X}_{\mathrm{v}}$, we can decompose $\mathbf{H}_t$ as follows:

$$\mathbf{H}_v = \mathbf{M}_v - \mathbf{M}_v\mathbf{Z}_{1,v}(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v})^{-1}\mathbf{Z}_{1,v}'\mathbf{M}_v$$

so that $h_t = m_t - \hat{z}_{1,t}'\,(Z_{1,u(t)}'M_vZ_{1,u(t)})^{-1}\hat{z}_{1,t}$. The second term is bounded as follows

$$\hat{z}_{1,r(v,t)}'\,(Z_{1,v}'M_vZ_{1,v})^{-1}\hat{z}_{1,r(v,t)} \leq \frac{\|\hat{z}_{1,r(v,t)}\|^2}{\lambda_{\min}(Z_{1,v}'M_vZ_{1,v})} = \frac{1}{\bar{T}}\,\frac{\|\hat{z}_{1,r(v,t)}\|^2}{\lambda_{\min}(Z_{1,v}'M_vZ_{1,v}/\bar{T})},$$

By Lemma 3, $\lambda_{\min}(Z_{1,v}'M_vZ_{1,v}/\bar{T}) \geq 1 - o_p(1)$ and $\max_t \|\hat{z}_{1,r(v,t)}\|^2/\bar{T} = o_p(1)$ so

$$\max_t \hat{z}_{1,r(v,t)}'\,(Z_{1,v}'M_vZ_{1,v})^{-1}\hat{z}_{1,r(v,t)} \leq \bar{T}^{-1}\max_t \|\hat{z}_{1,r(v,t)}\|^2 \cdot O_p(1) = o_p(1).$$

Hence $\min_t h_t > c - o_p(1)$ for some $c > 0$, so that the probability that $\max_t \frac{1}{h_t} < 2/c$ goes to 1. □

Theorem 4. *Under Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, Bounded Cross-Leverage, Bounded Fitted Values *and* Leave-One-Out Conditioning, $\frac{\acute{\Sigma}}{\Sigma} \xrightarrow{p} 1$.

Proof.

$$\begin{aligned}\acute{\Sigma} =& r' \sum_v \left(\mathbf{X}_v'\mathbf{X}_v\right)^{-1} \mathbf{X}_v' \acute{\mathbf{\Omega}}_v^2 \mathbf{X}_v \left(\mathbf{X}_v'\mathbf{X}_v\right)^{-1} r \\ =& \sum_v r'\Theta_v \left(\mathbf{Z}_v'\mathbf{Z}_v\right)^{-1} \mathbf{Z}_v' \acute{\mathbf{\Omega}}_v^2 \mathbf{Z}_v \left(\mathbf{Z}_v'\mathbf{Z}_v\right)^{-1} \Theta_v' r \\ =& \sum_v r_v' \left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} \mathbf{Z}_{1,v}'\mathbf{M}_v \acute{\mathbf{\Omega}}_v^2 \mathbf{M}_v\mathbf{Z}_{1,v} \left(\mathbf{Z}_{1,v}'\mathbf{M}_v\mathbf{Z}_{1,v}\right)^{-1} r_v\end{aligned}$$

where $\acute{\mathbf{\Omega}}_v^2 = \operatorname{diag}\left(\left(y_{r(v,t)} \frac{\hat{\varepsilon}_{r(v,t)}}{\mathbf{H}_{v,tt}}\right)_{t=1,\dots,\bar{T}}\right)$.

Let

$$\acute{W}_v := \frac{1}{\bar{T}} \sum_{t=1}^{\bar{T}} \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' \frac{y_{r(v,t)} \hat{\varepsilon}_{r(v,t)}}{\mathbf{H}_{v,tt}}, \qquad W_v := \frac{1}{\bar{T}} Z_{1,v}' M_v \Omega_v^2 M_v Z_{1,v} = \frac{1}{\bar{T}} \sum_{t=1}^{\bar{T}} \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' \, \sigma_{r(v,t)}^2,$$

(the meat of Σ and $\acute{\Sigma}$, respectively). We show that

$$\left\|\acute{W}_v - W_v\right\|_2 \xrightarrow{p} 0 \qquad \text{for each } v \in \{1, \dots, h+1\}; \tag{34}$$

as $\acute{W}_v$ and W_v are both normalised by $\bar{T}$, this is condition (26) of Lemma 11, and the rest of the proof follows from that lemma.

$$\acute{W}_v - W_v = \frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' \left(\frac{y_{r(v,t)} \hat{\varepsilon}_{r(v,t)}}{\mathbf{H}_{v,tt}} - \sigma_{r(v,t)}^2 \right).$$

Throughout, set $\acute{u}_t := \hat{\varepsilon}_{r(v,t)} / \mathbf{H}_{v,tt}$, so that the summand becomes $\hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' (y_t \acute{u}_t - \sigma_t^2)$. Adding and subtracting $\varepsilon_{r(v,t)}^2$,

$$\frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' (y_t \acute{u}_t - \sigma_t^2) = \frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' (\varepsilon_{r(v,t)}^2 - \sigma_t^2) + \frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' (y_t \acute{u}_t - \varepsilon_{r(v,t)}^2). \tag{35}$$

First term of (35)**.** Conditionally on X_v, the $\varepsilon_{r(v,t)}^2 - \sigma_t^2$ are mean zero and uncorrelated across t within the subsample (Assumption **??**, the normalized errors being iid). Consider the (k, k') entry; its conditional second moment is

$$\frac{1}{\bar{T}^2} \sum_{t,s} \hat{z}_{t,k} \hat{z}_{t,k'} \hat{z}_{s,k} \hat{z}_{s,k'} \, \mathbb{E}\left[(\varepsilon_{r(v,t)}^2 - \sigma_t^2)(\varepsilon_s^2 - \sigma_s^2) \;\middle|\; X_v \right] = \frac{1}{\bar{T}^2} \sum_t \hat{z}_{t,k}^2 \hat{z}_{t,k'}^2 \, \operatorname{Var}(\varepsilon_{r(v,t)}^2 \mid X_v),$$

which by $\operatorname{Var}(\varepsilon_{r(v,t)}^2 \mid X_v) \le \mathbb{E}[\varepsilon_{r(v,t)}^4] \le C^2$ (Assumption **??**) and $\hat{z}_{t,k}^2 \hat{z}_{t,k'}^2 \le \|\hat{z}_{1,r(v,t)}\|^4$ is at most

$$\frac{C^2}{\bar{T}} \cdot \frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^4 = \frac{C^2}{\bar{T}} \, O_p(1) = o_p(1)$$

by Lemma 10. Markov's inequality applied entrywise (the matrices are $(h+1) \times (h+1)$, a fixed dimension) gives $\left\| \frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}_{1,r(v,t)}' (\varepsilon_{r(v,t)}^2 - \sigma_t^2) \right\|_2 = o_p(1)$.

Decomposition of the second term of (35)**.** Define the full-annihilator-scaled matrix $(\mathcal{A}_v)_{tj} := \mathbf{H}_{v,tj}/\mathbf{H}_{v,tt}$. Since $y_t = \mu_t + \varepsilon_{r(v,t)}$ with $\mu_t = \beta' x_{r(v,t)}$, and $\hat{\varepsilon}_{r(v,t)} = \sum_j (\mathbf{H}_v)_{tj}\varepsilon_{r(v,j)}$ with, we have $\acute{u}_t = \sum_j (\mathcal{A}_v)_{tj}\varepsilon_{r(v,j)}$ and therefore

$$y_t \acute{u}_t - \varepsilon^2_{r(v,t)} = \varepsilon_{r(v,t)} \sum_{j\neq t} (\mathcal{A}_v)_{tj}\varepsilon_{r(v,j)} + \big((\mathcal{A}_v)_{tt} - 1\big)\varepsilon^2_{r(v,t)} + \mu_t \sum_j (\mathcal{A}_v)_{tj}\varepsilon_{r(v,j)}. \tag{36}$$

We first note that the middle term is zero because $(\mathcal{A}_v)_{tt} = \frac{\mathbf{H}_{v,tt}}{\mathbf{H}_{v,tt}} = 1$. Substituting what remains of (36) into the second term of (35) produces two matrix averages, which we denote T_1 and T_2 and bound in turn. We repeatedly use the relation

$$\sum_j (\mathcal{A}_v)^2_{tj} = \sum_j (\mathbf{H}_v)^2_{tj}/\mathbf{H}^2_{v,tt} = (\mathbf{H}_v)_{tt}/\mathbf{H}^2_{v,tt} \leq \mathbf{H}^{-2}_{v,tt} = O_p(1)$$

, which follows from idempotency of $\mathbf{H}_v$ ($\sum_j (\mathbf{H}_v)^2_{tj} = (\mathbf{H}_v)_{tt}$) and $\mathbf{H}_{v,tt} \leq 1$, together with $1/\min_t \mathbf{H}_{v,tt} = O_p(1)$ (Lemma 14).

T_1**: the quadratic cross term.** We bound the conditional variance of $T_1 = \frac{1}{\bar{T}} \sum_t \sum_{j\neq t} \hat{z}_{1,r(v,t)} \hat{z}'_{1,r(v,t)} \varepsilon_{r(v,t}$ By the uncorrelatedness, the only surviving cross-products in $\mathbb{E}[\,T_1^2 \mid X_v]$ when taking the expectation of the squared sum are the "diagonal" pairings $(t_1, j_1) = (t_2, j_2)$ and the "transposed" pairings $(t_1, j_1) = (j_2, t_2)$, so the conditional second moment of any fixed entry is bounded by

$$\frac{C^2}{\bar{T}^2} \sum_t \sum_{j\neq t} \|\hat{z}_{1,r(v,t)}\|^4 (\mathcal{A}_v)^2_{tj} + \frac{C^2}{\bar{T}^2} \sum_t \sum_{j\neq t} \|\hat{z}_{1,r(v,t)}\|^2 \|\hat{z}_{1,r(v,j)}\|^2 |(\mathcal{A}_v)_{tj}(\mathcal{A}_v)_{jt}|,$$

using $\mathbb{E}[\varepsilon^2_{r(v,t)}\varepsilon^2_j \mid X_v] \leq C^2$. For the first sum, $\sum_{j\neq t} (\mathcal{A}_v)^2_{tj} \leq \mathbf{H}^{-2}_{v,tt}$, so it is at most

$$\frac{C^2}{\bar{T}} \Big(\max_t \frac{1}{\mathbf{H}^2_{v,tt}}\Big) \cdot \frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^4 = \frac{1}{\bar{T}}\, O_p(1)\, O_p(1) = o_p(1),$$

by $1/\min_t \mathbf{H}_{v,tt} = O_p(1)$ and Lemma 10. For the second, we proceed similarly (using $(\mathcal{A}_v)_{tj}(\mathcal{A}_v)_{jt} = \frac{\mathbf{H}^2_{v,tj}}{\mathbf{M}_{v,tt}\mathbf{M}_{v,jj}}$)

$$\frac{C^2}{\bar{T}^2} \sum_t \sum_{j\neq t} \|\hat{z}_{1,r(v,t)}\|^2 \|\hat{z}_{1,r(v,j)}\|^2 |(\mathcal{A}_v)_{tj}(\mathcal{A}_v)_{jt}| \leq \max \frac{1}{\bar{T}} \max_t \|\hat{z}_{1,r(v,t)}\|^2 \min_t \mathbf{H}^2_{v,tt} \frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^2 = \quad o_p(1)$$

(using Lemmas 10 and 13.)

T_2**: the leverage-correction term.** $T_2 = \frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}'_{1,r(v,t)} \big((\mathcal{A}_v)_{tt} - 1\big)\varepsilon^2_{r(v,t)}$. Since $(\mathcal{A}_v)_{tt} = (\mathbf{H}_v)_{tt}/\mathbf{H}_{v,tt}$ and $\mathbf{H}_{v,tt} \leq \mathbf{H}_{v,tt}$, we have $|(\mathcal{A}_v)_{tt} - 1| = |\mathbf{H}_{v,tt} - \mathbf{H}_{v,tt}|/\mathbf{H}_{v,tt} = \hat{z}_{1,r(v,t)}(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v})^{-1}\hat{z}_{1,r(v,t)}/\mathbf{H}_{v,tt}$. $\max_t \hat{z}_{1,r(v,t)}(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v})^{-1}\hat{z}_{1,r(v,t)} = o_p(1)$ (as we saw above) and $1/\min_t \mathbf{H}_{v,tt} = O_p(1)$,

$$\begin{aligned}\max_t \big|(\mathcal{A}_v)_{tt} - 1\big| =& \max_t \left|\frac{\mathbf{H}_{v,tt} - \mathbf{H}_{v,tt}}{\mathbf{H}_{v,tt}}\right| \\ =& o_p(1).\end{aligned}$$

Therefore, taking conditional expectations and using $\mathbb{E}[\varepsilon_{r(v,t)}^2 \mid X_v] = \sigma_t^2 \le C$,

$$\|T_2\|_2 \le \Big(\max_t |(\mathcal{A}_v)_{tt} - 1|\Big) \cdot \frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^2 \varepsilon_{r(v,t)}^2.$$

The average $\frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^2 \varepsilon_{r(v,t)}^2$ is $O_p(1)$: its conditional mean is $\frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^2 \sigma_t^2 \le C \cdot \frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^2 = O_p(1)$. Hence $\|T_2\|_2 = o_p(1) \cdot O_p(1) = o_p(1)$.

T_3: the μ term. $T_3 = \frac{1}{\bar{T}} \sum_t \hat{z}_{1,r(v,t)} \hat{z}'_{1,r(v,t)} \mu_t \sum_j (\mathcal{A}_v)_{tj} \varepsilon_{r(v,j)}$ has conditional mean zero (again $\mathbb{E}[\varepsilon_{r(v,j)} \mid X_v] = 0$). Fix the (k, k') entry. Its conditional variance is

$$\mathrm{Var}\Big(\frac{1}{\bar{T}} \sum_t \hat{z}_{t,k} \hat{z}_{t,k'} \mu_t \sum_j (\mathcal{A}_v)_{tj} \varepsilon_{r(v,j)} \;\Big|\; X_v\Big) = \frac{1}{\bar{T}^2} \sum_j \sigma_{r(v,j)}^2 \Big(\sum_t \hat{z}_{t,k} \hat{z}_{t,k'} \mu_t (\mathcal{A}_v)_{tj}\Big)^2.$$

Bound $\sigma_{r(v,j)}^2 \le C$ and use the quadratic-form structure: writing $a_t := \hat{z}_{t,k} \hat{z}_{t,k'} \mu_t$, the inner double sum is

$$\sum_j (\sum_t a_t (\mathcal{A}_v)_{tj})^2 = \sum_{j,t,s} a_t a_s (\mathcal{A}_v)_{tj} (\mathcal{A}_v)_{sj} = \sum_{t,s} a_t a_s \sum_j (\mathcal{A}_v)_{tj} (\mathcal{A}_v)_{sj} = \sum_{t,s} a_t a_s \frac{(\mathbf{H}_v)_{ts}}{\mathbf{H}_{v,tt} M_{v,ss}}.$$

Since $\mathbf{H}_v$ is a projection ($\sum_{t,s} a_t (\mathbf{H}_v)_{ts} a_s \le \sum_t a_t^2$),

$$\sum_{t,s} a_t a_s \frac{(\mathbf{H}_v)_{ts}}{\mathbf{H}_{v,tt} M_{v,ss}} \le \Big(\max_t \frac{1}{\mathbf{H}_{v,tt}^2}\Big) \sum_{t,s} a_t (\mathbf{H}_v)_{ts} a_s \le \Big(\max_t \frac{1}{\mathbf{H}_{v,tt}^2}\Big) \sum_t a_t^2,$$

Thus the conditional variance is at most

$$\frac{C}{\bar{T}^2} \Big(\max_t \frac{1}{\mathbf{H}_{v,tt}^2}\Big) \sum_t a_t^2 = \frac{C}{\bar{T}} O_p(1) \cdot \frac{1}{\bar{T}} \sum_t \hat{z}_{t,k}^2 \hat{z}_{t,k'}^2 \mu_t^2.$$

Finally $\frac{1}{\bar{T}^2} \sum_t \hat{z}_{t,k}^2 \hat{z}_{t,k'}^2 \mu_t^2 \le \frac{\max_t \mu_t^2}{\bar{T}} \cdot \frac{1}{\bar{T}} \sum_t \|\hat{z}_{1,r(v,t)}\|^4 = o_p(1) \cdot O_p(1)$ by Assumption *Bounded Fitted Values* and Lemma 10 under assumption *Bounded Cross-Leverage*.

The conditional variance of each part (T_1 to T_3) vanishes, implying that the second term in (35) is $o_p(1)$ as well so that $\|R_{2,v}\|_2 = o_p(1)$. We conclude that $\|\acute{W}_v - W_v\|_2 \le \|R_{1,v}\|_2 + \|R_{2,v}\|_2 = o_p(1)$. (And because $\lambda_{\min}(W_v)$ is bounded below, that means that $W_v^{-1} \acute{W}_v \xrightarrow{p} \mathbf{I}_{h+1}$.)

This establishes (34). Since $\acute{\Sigma}$ is of the sandwich form (**??**), Lemma 11 now yields $\frac{\acute{\Sigma}}{\Sigma} \xrightarrow{p} 1$. □

Lemma 15. *Under Assumptions* Feedback Decomposition *and* Sub-One-Half Leverage*, for every t in subsample v,*

$$M_{v,tt} \;\ge\; 1 - \tilde{x}_t' (\tilde{\mathbf{X}}_v' \tilde{\mathbf{X}}_v)^{-1} \tilde{x}_t \;\ge\; \frac{1}{2} + c_\kappa,$$

where $\tilde{z}_{2,t} = \boldsymbol{\Theta}'_{v,2} \tilde{x}_t$, deterministically.

Proof. The first inequality follows from the fact that $\mathrm{col}\{\mathbf{Z}_{2,v}\} \subseteq \mathrm{col}\{\mathbf{Z}_v\}$. The second inequality is Assumption *Sub-One-Half Leverage*. □

Lemma 16. *Under Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, Bounded Cross-Leverage *and* Sub-One-Half Leverage,

$$\mathbb{P}\left[\min_{1\le t\le \bar{T}} H_{v,tt} \ge \frac{1}{2} + \frac{c_\kappa}{2}\right] \to 1.$$

Proof. By the Frisch–Waugh–Lovell decomposition of the full leverage into its nuisance- and contrast-block contributions (as in the proof of Lemma 14),

$$H_{v,tt} = M_{v,tt} - \hat{z}'_{1,r(v,t)}(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v})^{-1}\hat{z}_{1,r(v,t)},$$

where $\hat{z}_{1,r(v,t)}$ is row t of $\mathbf{M}_v\mathbf{Z}_{1,v}$.

$$\max_t \hat{z}'_{1,r(v,t)}(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v})^{-1}\hat{z}_{1,r(v,t)} \le \frac{\max_t \|\hat{z}_{1,r(v,t)}\|^2}{\bar{T}\,\lambda_{\min}\left(\mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{Z}_{1,v}/\bar{T}\right)} = \frac{o_p(\bar{T})}{\bar{T}\,(1-o_p(1))}.$$

using $\max_t \|\hat{z}_{1,r(v,t)}\|^2 = o_p(\bar{T})$ from Lemma 13 – which requires only Assumptions *Bounded Cross-Leverage* and *No Influential Observations*, not vanishing leverage – and Lemma 3. Combining with $M_{v,tt} \ge \frac{1}{2} + c_\kappa$ (Lemma 15) gives the claim. □

Lemma 17. *On the event* $\{\min_t H_{v,tt} \ge \frac{1}{2} + \frac{c_\kappa}{2}\}$ *of Lemma 16,* $\mathbf{H}_v \odot \mathbf{H}_v$ *is invertible and*

$$\|\boldsymbol{\kappa}_v\|_2 \;\le\; \frac{2}{c_\kappa}.$$

In particular, $\boldsymbol{\kappa}_v$ *is well defined with probability approaching one and* $\|\boldsymbol{\kappa}_v\|_2 = O_p(1)$. *(Off this vanishing-probability event, define* $\hat{\Sigma}^{CJN}$ *arbitrarily.)*

Proof. Since $\mathbf{H}$ is symmetric and idempotent, $\sum_s \mathbf{H}_{ts}^2 = (\mathbf{H}^2)_{tt} = \mathbf{H}_{tt}$, so the off-diagonal row sums are $\sum_{s\neq t} G_{ts} = \mathbf{H}_{tt} - \mathbf{H}_{tt}^2$. The dominance margin in row t is therefore

$$G_{tt} - \sum_{s\neq t} G_{ts} = \mathbf{H}_{tt}^2 - \big(\mathbf{H}_{tt} - \mathbf{H}_{tt}^2\big) = \mathbf{H}_{tt}\big(2\mathbf{H}_{tt} - 1\big) \;\ge\; \Big(\tfrac{1}{2} + c\Big)2c \;\ge\; c,$$

which is positive precisely when $\mathbf{H}_{tt} > 1/2$. As G is symmetric, Lemma 27 gives $\|G^{-1}\|_2 \le 1/c$. □

Theorem 3. *Under Assumptions* Feedback Decomposition, Bounded Eigenvalues, Regular Errors, Moderate-Dimensional Regime, Bounded Cross-Leverage *and* Sub-One-Half Leverage, $\mathbf{H}_v \odot \mathbf{H}_v$ *is invertible with probability approaching* 1 *and when choosing* $\kappa_v = (\mathbf{H}_v \odot \mathbf{H}_v)^{-1}$, $\frac{\hat{\Sigma}}{\Sigma} \xrightarrow{p} 1$.

Proof. Let $\hat{W}_v := \mathbf{Z}'_{1,v}\mathbf{M}_v\hat{\mathbf{\Omega}}_v\mathbf{M}_v\mathbf{Z}_{1,v}$, where $\hat{\mathbf{\Omega}}_v = \mathrm{diag}\,(\boldsymbol{\kappa}_v\hat{\varepsilon}_v^2)$, for $\hat{\varepsilon}_v^2$, the vector or squared residuals of subsample v. We show that

$$\left\|\frac{1}{T}(\hat{W}_v - \mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{\Omega}_v^2\mathbf{M}_v\mathbf{Z}_{1,v})\right\|_2 \xrightarrow{p} 0. \tag{37}$$

We assume $\mathbf{H} \odot \mathbf{H}$ to be invertible, the probabiltiy of which goes to 1 under Assumption *Sub-One-Half Leverage* given Lemma 16. First, as explained in the main text, $\mathbb{E}\left[\hat{\mathbf{\Omega}}_v | \mathbf{X}_v\right] = \Omega_v$ so that:

$$\mathbb{E}\left[\hat{W}_v \mid \mathbf{X}_v\right] = \mathbf{Z}'_{1,v}\mathbf{M}_v\mathbf{\Omega}\mathbf{M}_v\mathbf{Z}_{1,v}$$

As this matrix is of fixed dimension, it suffices to show that each entry concentrates around its conditional mean at rate $o_p(\bar{T})$. Fix $k, l \in \{1, \ldots, h+1\}$ and let w be the $\bar{T}$-vector with entries $w_t := \hat{z}_{1,r(v,t),k}\, \hat{z}_{1,r(v,t),l}$, so that, using the symmetry of $\boldsymbol{\kappa}_v$,

$$\big(\hat{W}_v\big)_{kl} = \sum_t w_t \big(\boldsymbol{\kappa}_v \hat{e}_v\big)_t = b'\hat{e}_v = \sum_j b_j\, \hat{\varepsilon}^2_{r(v,j)} = \varepsilon^{(v)\prime}\, \tilde{\mathbf{B}}\, \varepsilon^{(v)}, \qquad b := \boldsymbol{\kappa}_v w, \quad \tilde{\mathbf{B}} := \mathbf{H}_v\, diag(b)\, \mathbf{H}_v,$$

where the last step substituted $\hat{\varepsilon}^{(v)} = \mathbf{H}_v \varepsilon^{(v)}$. Both b and $\mathbf{H}_v$ are $\mathbf{X}_v$-measurable, and $\varepsilon^{(v)}$ is independent of $\mathbf{X}_v$ with independent entries, variances in $[c, C]$ and fourth moments bounded by C^2 (Assumption *Regular Errors*). Two bounds:

(a)

$$\begin{aligned}
\|w\|^2 &= \sum_t \left(\hat{z}_{1,r(v,t),k} \hat{z}_{1,r(v,t),l}\right)^2 \\
&\leq \sqrt{\sum_t \hat{z}^4_{1,r(v,t),k} \sum_t \hat{z}^4_{1,r(v,t),k}} \\
&\leq \sum_t \|\hat{z}_{1,r(v,t)}\|^4 = O_p(\bar{T})
\end{aligned}$$

by Lemma 10 under Assumption *Bounded Cross-Leverage*.

(b) $\|b\| \leq \|\boldsymbol{\kappa}_v\|_2 \|w\| \leq \frac{2}{c_\kappa}\|w\| = O_p(\bar{T}^{1/2})$, by Lemma 17 and (a).

By Lemma 2(ii) applied conditionally on $\mathbf{X}_v$ to the quadratic form $\varepsilon^{(v)\prime}\tilde{\mathbf{B}}\varepsilon^{(v)}$,

$$\big(\hat{W}_v\big)_{kl} - \mathbb{E}\left[\big(\hat{W}_v\big)_{kl} \mid \mathbf{X}_v\right] = O_p\big(\left\|\tilde{\mathbf{B}}\right\|_F\big), \qquad \left\|\tilde{\mathbf{B}}\right\|_F \leq \|\mathbf{H}_v\|_2^2\, \|diag(b)\|_F = \|b\| = O_p(\bar{T}^{1/2}),$$

using $\|\mathbf{H}_v\|_2 \leq 1$ and (b). This proves (37), which is condition (26) of Lemma 11 for the meat $\hat{\mathbf{\Psi}}_v = \hat{W}_v$, thus $\frac{\tilde{\Sigma}}{\Sigma} \xrightarrow{p} 1$. □

B Matrix basics

B.1 Matrix norms

A matrix norm $\|\cdot\|$ on $\mathbb{R}^{m\times n}$ satisfies:

- **Non-negativity:** $\|A\| \geq 0$, and $\|A\| = 0 \iff A = 0$
- **Homogeneity:** $\|\alpha A\| = |\alpha|\|A\|$ for all $\alpha \in \mathbb{R}$
- **Triangle inequality:** $\|A + B\| \leq \|A\| + \|B\|$

If the norm is *submultiplicative*:

$$\|AB\| \leq \|A\|\, \|B\|$$

B.1.1 The spectral norm

The spectral (operator) norm is defined as:

$$\|A\|_2 = \sup_{\|x\|_2=1} \|Ax\|_2$$

Properties

- $$\|A\|_2 = \sqrt{\lambda_{\max}(A'A)}$$

  Thus, if A is symmetric, $\|A\|_2 = \lambda_{\max}(A)$.
- $\|A\|_2 = \|A'\|_2$
- The spectral norm is submultiplicative

B.1.2 Induced norms and the maximum row- and column-sum norms

Every vector norm $\|\cdot\|$ induces a matrix norm via

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|},$$

so that $\|Ax\| \leq \|A\| \|x\|$ for all x by construction; this immediately makes every induced norm submultiplicative, since $\|ABx\| \leq \|A\| \|Bx\| \leq \|A\| \|B\| \|x\|$. The spectral norm is the norm induced by the Euclidean vector norm. Two further induced norms are used in this paper.

Lemma 18 (Maximum absolute row sum)**.** *The matrix norm induced by the max-norm* $\|x\|_\infty = \max_i |x_i|$ *is the maximum absolute row sum,*

$$\|A\|_\infty = \max_i \sum_j |a_{ij}|.$$

Proof. For any x with $\|x\|_\infty \leq 1$,

$$\|Ax\|_\infty = \max_i \left| \sum_j a_{ij} x_j \right| \leq \max_i \sum_j |a_{ij}| \, |x_j| \leq \max_i \sum_j |a_{ij}|,$$

so the induced norm is at most the maximum absolute row sum. Conversely, assume $A \neq 0$, let $i^\star$ attain the maximum and set $x_j = \operatorname{sgn}(a_{i^\star j})$, so that $\|x\|_\infty \leq 1$ and the $i^\star$-th entry of Ax equals $\sum_j |a_{i^\star j}|$. Hence $\|Ax\|_\infty \geq \sum_j |a_{i^\star j}|$, and the two bounds coincide. The absolute values are essential: without the sign adjustment, entries of opposite sign within a row would cancel. □

Correspondingly, the maximum absolute column sum is

$$\|A\|_1 = \max_j \sum_i |a_{ij}| = \|A'\|_\infty,$$

the second equality holding because the rows of A' are the columns of A; an argument analogous to that of Lemma 18 shows that this is the norm induced by the vector norm $\|x\|_1 = \sum_i |x_i|$. Note the mnemonic hazard: the 1-norm sums down columns, the ∞-norm across rows.

Lemma 19 (Eigenvalues are bounded by any induced norm)**.** *Let $\|\cdot\|$ be any induced matrix norm and let λ be any eigenvalue of $A \in \mathbb{R}^{n\times n}$. Then $|\lambda| \le \|A\|$. In particular the spectral radius satisfies $\rho(A) := \max_i |\lambda_i(A)| \le \|A\|$.*

Proof. Let $x \neq 0$ satisfy $Ax = \lambda x$. Then

$$|\lambda|\,\|x\| = \|\lambda x\| = \|Ax\| \le \|A\|\,\|x\|,$$

and dividing by $\|x\| > 0$ gives the claim. Only the defining property $\|Ax\| \le \|A\|\,\|x\|$ of an induced norm is used, so the bound holds for $\|\cdot\|_2$, $\|\cdot\|_1$ and $\|\cdot\|_\infty$ alike. (Eigenvalues and eigenvectors of a general square matrix may be complex, in which case the norm is read on $\mathbb{C}^n$; the formula of Lemma 18 is unchanged there, with the sign choice replaced by the unit-modulus phases $x_j = \overline{a_{i^\star j}}/|a_{i^\star j}|$.) □

Lemma 20 (Interpolation between the row- and column-sum norms)**.** *For any $A \in \mathbb{R}^{n\times n}$,*

$$\|A\|_2 \le \sqrt{\|A\|_1\,\|A\|_\infty}.$$

If A is symmetric, then $\|A\|_1 = \|A\|_\infty$ and hence $\|A\|_2 \le \|A\|_\infty$.

Proof. $A'A$ is symmetric positive semidefinite, so its eigenvalues are real and non-negative and $\|A\|_2^2 = \lambda_{\max}(A'A) = \rho(A'A)$. Applying Lemma 19 with the maximum-row-sum norm and then submultiplicativity,

$$\|A\|_2^2 = \rho(A'A) \le \|A'A\|_\infty \le \|A'\|_\infty\,\|A\|_\infty = \|A\|_1\,\|A\|_\infty.$$

If $A = A'$ its rows and columns coincide, so $\|A\|_1 = \|A\|_\infty$ and the bound reads $\|A\|_2 \le \|A\|_\infty$. □

This is the standard route for transferring a bound obtained in the max-norm to the spectral norm: the two norms are in general not comparable, but for symmetric matrices the interpolation inequality collapses and the row-sum bound carries over at no cost.

B.1.3 Frobenius Norm

The Frobenius norm is defined as:

$$\|A\|_F = \left(\sum_{i,j} A_{ij}^2\right)^{1/2}$$

Equivalently:

$$\|A\|_F = \sqrt{\mathrm{tr}(A'A)}$$

B.1.4 Inequalities Between Norms

- Relation between spectral and Frobenius norms:

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{\operatorname{rank}(A)}\, \|A\|_2$$

- Submultiplicativity:

$$\|AB\|_2 \leq \|A\|_2\, \|B\|_2, \quad \|AB\|_F \leq \|A\|_F\, \|B\|_F$$

(In fact, for the Frobenius norm, a tighter bound is available: $\|AB\|_F \leq \|A\|_2\, \|B\|_F$.)

- Trace inequality (useful for products):

$$|\operatorname{tr}(A'B)| \leq \|A\|_F\, \|B\|_F$$

- Trace-norm inequality:

Lemma 21 (Trace-norm inequality)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite and let $B \in \mathbb{R}^{n\times n}$ be arbitrary. Then*

$$|\operatorname{tr}(AB)| \leq \|B\|_2\, \operatorname{tr}(A),$$

where $\|B\|_2$ denotes the spectral norm of B.

Proof. Since A is symmetric positive semidefinite, it admits an eigenvalue decomposition

$$A = \sum_{i=1}^{n} \lambda_i u_i u_i',$$

where $\lambda_i \geq 0$ for all i, and $\{u_i\}_{i=1}^n$ is an orthonormal basis of $\mathbb{R}^n$.

Using linearity and cyclicity of the trace, we obtain

$$\operatorname{tr}(AB) = \sum_{i=1}^{n} \lambda_i \operatorname{tr}(u_i u_i' B) = \sum_{i=1}^{n} \lambda_i\, u_i' B u_i.$$

Hence:

$$|\operatorname{tr}(AB)| \leq \sum_{i=1}^{n} \lambda_i\, u_i' B u_i.$$

For each i, we bound the quadratic form using the spectral norm. Note that

$$u_i' B u_i = u_i' \left(\tfrac{B+B'}{2}\right) u_i,$$

and since $\left\|\frac{B+B'}{2}\right\|_2 \leq \|B\|_2$ (triangle inequality and the fact that BB' and $B'B$ have the same eigenvalues), it follows that

$$|u_i' B u_i| \leq \|B\|_2 \|u_i\|^2 = \|B\|_2.$$

Therefore,

$$|\operatorname{tr}(AB)| \leq \sum_{i=1}^{n} \lambda_i \|B\|_2 = \|B\|_2 \sum_{i=1}^{n} \lambda_i = \|B\|_2 \operatorname{tr}(A),$$

which proves the claim. □

Lemma 22. *Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite and let $\mathbf{B} \in \mathbb{R}^{n\times n}$ be symmetric. Then*

$$\operatorname{tr}(\mathbf{AB}) \geq \lambda_{\min}(\mathbf{B}) \operatorname{tr}(\mathbf{A}).$$

Proof. Since $\mathbf{A}$ is symmetric PSD, it admits the spectral decomposition $\mathbf{A} = \sum_{i=1}^{n} \lambda_i(\mathbf{A})\, u_i u_i'$, where $\lambda_i(\mathbf{A}) \geq 0$ and $\{u_i\}_{i=1}^{n}$ is an orthonormal basis of $\mathbb{R}^n$. Therefore

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}\left(\sum_{i=1}^{n} \lambda_i(\mathbf{A})\, u_i u_i' \mathbf{B}\right) = \sum_{i=1}^{n} \lambda_i(\mathbf{A})\, u_i' \mathbf{B}\, u_i.$$

Since $\mathbf{B}$ is symmetric, the Rayleigh quotient satisfies $u_i' \mathbf{B}\, u_i \geq \lambda_{\min}(\mathbf{B}) \|u_i\|^2 = \lambda_{\min}(\mathbf{B})$ for each i. Hence

$$\operatorname{tr}(\mathbf{AB}) \geq \lambda_{\min}(\mathbf{B}) \sum_{i=1}^{n} \lambda_i(\mathbf{A}) = \lambda_{\min}(\mathbf{B}) \operatorname{tr}(\mathbf{A}). \qquad \square$$

B.1.5 Bounds on columns and rows

Let $A = [a_1, \ldots, a_n]$ with columns a_j.

- Largest column (Euclidean norm) is bounded by:

$$\max_j \|a_j\|_2 \leq \|A\|_2$$

- Similarly for rows:

$$\max_i \|A_{i\cdot}\|_2 \leq \|A\|_2$$

Spectral norm and Frobenius norm of projection matrices

Let P be an orthogonal projection matrix ($P^2 = P$, $P' = P$).

- Eigenvalues are 0 or 1 hence spectral norm:

$$\|P\|_2 = 1$$

- Frobenius norm:

$$\|P\|_F = \sqrt{\operatorname{tr}(P)} = \sqrt{\operatorname{rank}(P)}$$

B.2 Eigenvalues

- **Weyl's inequality.** Let A and B we hermitian matrices of dimension p. Let $\lambda_1(Q), \ldots, \lambda_n(Q)$ denote the eigenvalues of an $n \times n$ matrix Q in ascending order. Then for i, j with $1 \leq i + j - 1 \leq n$:

$$\lambda_i(A) + \lambda_j(B) \leq \lambda_{i+j-1}(A + B)$$

  It follows that

$$\lambda_{min}(A + B) \geq \lambda_{min}(A) + \lambda_{min}(B)$$

- For any real-valued matrix A, the non-zero eigenvalues of AA' and $A'A$ coincide. As a consequence $\|A\|_2 = \|A'\|_2$.

  Lemma 23. *Let $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{n\times m}$. Then*

  $$\lambda_{\min}(A'A)\lambda_{\min}(B'B) \leq \lambda_{\min}(A'B'BA)$$

  and

  $$\lambda_{\max}(A'B'BA) \leq \lambda_{\max}(A'A)\lambda_{\max}(B'B).$$

  Proof. For any vector x,

  $$x'A'B'BAx = (Ax)'(B'B)(Ax).$$

  Using the Rayleigh quotient bound,

  $$(Ax)'(B'B)(Ax) \geq \lambda_{\min}(B'B)\|Ax\|^2.$$

  Since

  $$\|Ax\|^2 = x'A'Ax \geq \lambda_{\min}(A'A)\|x\|^2,$$

  it follows that

  $$x'A'B'BAx \geq \lambda_{\min}(B'B)\lambda_{\min}(A'A)\|x\|^2.$$

  Taking the minimum over all $x \neq 0$ yields

  $$\lambda_{\min}(A'B'BA) \geq \lambda_{\min}(A'A)\lambda_{\min}(B'B).$$

  Similarly,

  $$x'A'B'BAx = (Ax)'(B'B)(Ax) \leq \lambda_{\max}(B'B)\|Ax\|^2$$

  and

  $$\|Ax\|^2 \leq \lambda_{\max}(A'A)\|x\|^2.$$

  Therefore

  $$x'A'B'BAx \leq \lambda_{\max}(B'B)\lambda_{\max}(A'A)\|x\|^2.$$

  Taking the maximum over $x \neq 0$ gives

  $$\lambda_{\max}(A'B'BA) \leq \lambda_{\max}(A'A)\lambda_{\max}(B'B).$$

  □

Lemma 24. *Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let B be a principal submatrix of A. Then*

$$\lambda_{\min}(A) \leq \lambda_{\min}(B) \leq \lambda_{\max}(B) \leq \lambda_{\max}(A).$$

Proof. Without loss of generality, write

$$A = \begin{pmatrix} B & C \\ C' & D \end{pmatrix},$$

where B is a $k \times k$ principal submatrix of A.

For any $x \in \mathbb{R}^k$,

$$x'Bx = \begin{pmatrix} x \\ 0 \end{pmatrix}' A \begin{pmatrix} x \\ 0 \end{pmatrix}.$$

Hence,

$$x'Bx \geq \lambda_{\min}(A)\|x\|^2.$$

Taking the minimum over all $x \neq 0$ yields

$$\lambda_{\min}(B) \geq \lambda_{\min}(A).$$

Similarly,

$$x'Bx \leq \lambda_{\max}(A)\|x\|^2,$$

and therefore

$$\lambda_{\max}(B) \leq \lambda_{\max}(A).$$

Combining the two inequalities gives

$$\lambda_{\min}(A) \leq \lambda_{\min}(B) \leq \lambda_{\max}(B) \leq \lambda_{\max}(A).$$

□

B.3 Diagonal dominance

A matrix $A \in \mathbb{R}^{n \times n}$ is *strictly diagonally dominant* if each diagonal entry exceeds, in absolute value, the sum of the absolute values of the remaining entries in its row:

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|, \qquad i = 1, \ldots, n.$$

We say that the dominance holds *with margin* $\alpha > 0$ if

$$|a_{ii}| - \sum_{j \neq i} |a_{ij}| \ \geq \ \alpha, \qquad i = 1, \ldots, n.$$

Lemma 25 (Gershgorin's circle theorem)**.** *Every eigenvalue λ of A lies in one of the discs*

$$D_i = \Big\{ z \ : \ |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \Big\}, \qquad i = 1, \ldots, n.$$

Proof. Let $Ax = \lambda x$ with $x \neq 0$ and choose i with $|x_i| = \max_j |x_j| > 0$. The i-th row of the eigenvalue equation reads $\sum_j a_{ij}x_j = \lambda x_i$, that is,

$$(\lambda - a_{ii})\, x_i = \sum_{j \neq i} a_{ij} x_j.$$

Taking absolute values and using $|x_j| \leq |x_i|$ for all j,

$$|\lambda - a_{ii}|\, |x_i| \;\leq\; \sum_{j \neq i} |a_{ij}|\, |x_j| \;\leq\; |x_i| \sum_{j \neq i} |a_{ij}|,$$

and dividing by $|x_i| > 0$ places λ in D_i. □

Gershgorin's theorem also gives a second proof of Lemma 19 for the max-norm, since every disc D_i is contained in the disc of radius $|a_{ii}| + \sum_{j \neq i} |a_{ij}| = \sum_j |a_{ij}| \leq \|A\|_\infty$ centred at the origin. The following two consequences are two further readings of the same picture: strict diagonal dominance says precisely that no Gershgorin disc reaches the origin.

Lemma 26 (Strict diagonal dominance implies invertibility)**.** *If A is strictly diagonally dominant, then A is invertible.*

Proof. Suppose $Ax = 0$ for some $x \neq 0$. Normalising, we may assume $\|x\|_\infty = 1$; pick t with $|x_t| = 1$. The t-th row gives $a_{tt}x_t = -\sum_{s \neq t} a_{ts}x_s$, so

$$|a_{tt}| \;\leq\; \sum_{s \neq t} |a_{ts}|\, |x_s| \;\leq\; \sum_{s \neq t} |a_{ts}|,$$

contradicting strict dominance. Hence the null space of A is trivial. Equivalently, by Lemma 25 every eigenvalue lies in a disc centred at a_{ii} of radius $\sum_{j \neq i} |a_{ij}| < |a_{ii}|$, and no such disc contains the origin, so 0 is not an eigenvalue of A. □

The following result is due to Varah (Varah (1975)).

Lemma 27 (Varah's bound)**.** *If A is strictly diagonally dominant with margin $\alpha > 0$, then*

$$\|A^{-1}\|_\infty \;\leq\; \frac{1}{\alpha}.$$

If in addition A is symmetric, then $\|A^{-1}\|_2 \leq 1/\alpha$.

Proof. Let x satisfy $\|x\|_\infty = 1$ and pick t with $|x_t| = 1$. Bounding $\|Ax\|_\infty$ below by the absolute value of its t-th entry,

$$\|Ax\|_\infty \;\geq\; \left| a_{tt}x_t + \sum_{s \neq t} a_{ts}x_s \right| \;\geq\; |a_{tt}| - \sum_{s \neq t} |a_{ts}| \;\geq\; \alpha,$$

using $|x_t| = 1$ and $|x_s| \leq 1$. By homogeneity, $\|Ax\|_\infty \geq \alpha \|x\|_\infty$ for every x. The matrix A is invertible by Lemma 26, so substituting $x = A^{-1}y$ gives $\|y\|_\infty \geq \alpha\, \|A^{-1}y\|_\infty$ for every y, i.e. $\|A^{-1}\|_\infty \leq 1/\alpha$. In words: a matrix that is diagonally dominant with margin α cannot shrink any vector in the max-norm by more than the factor α, so its inverse cannot stretch one by more than $1/\alpha$. If A is symmetric then so is A^{-1}, and Lemma 20 upgrades the bound to the spectral norm, $\|A^{-1}\|_2 \leq \|A^{-1}\|_\infty \leq 1/\alpha$. □

C Leverage simulation

This appendix provides simulation evidence on the maximum-leverage condition of Assumption *No Influential Observations*, under the DGP described in Section 4.

As noted in Section 3, the ordinary leverage $h_t := \tilde{x}_t'(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{x}_t$ satisfies $\max_t h_t \geq K/\bar{T}$, so it cannot vanish whenever the aspect ratio $K/\bar{T}$ does not vanish. However, for the purposes of asymptotic normality of the divide-and-conquer estimator only

$$\max_t \frac{1}{\sqrt{T}} \|A(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{x}_t\|$$

must vanish according to assumption *No Influential Observations*. We simulate this quantity using $A = (1, 0, 0, \ldots, 0)$ under the DGP used in section 4 with $K/T = 0.5$, $\rho = 0.9$, $\sigma_\eta = 0.2$, $\nu = 2.1$ and $\nu = 2.5$, respectively. We also include a the case where the VAR innovations generating $\tilde{x}_t$ are Gaussian. Recall that we normalize innovations to have the variance ($\sigma_\eta = 0.2$) for all choices of ν. We ignore subsampling. Table 3 reports the mean and standard deviation of $\max_t \frac{1}{\sqrt{T}} \|e_1'(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{x}_t\|$ across Monte Carlo replications, for a range of sample sizes. The simulations confirm that assumption *No Influential Observations* is realistic even when the $\tilde{x}_t$ are rather heavy-tailed.

Table 3: $\max_t |e_1'(\tilde{\mathbf{X}}_v'\tilde{\mathbf{X}}_v)^{-1}\tilde{x}_t|/\sqrt{T}$ under a diagonal VAR(1)

Sample size	Mean	Std. dev.
$\nu = 2.1$		
33	0.2289	0.0936
100	0.0522	0.0154
300	0.0125	0.0027
900	0.0033	0.0008
$\nu = 2.5$		
33	0.1291	0.0462
100	0.0290	0.0072
300	0.0070	0.0015
900	0.0018	0.0004
$\nu = \infty$		
33	0.0945	0.0260
100	0.0210	0.0036
300	0.0046	0.0006
900	0.0010	0.0001

Notes: $\tilde{x}_t$ follows a diagonal VAR(1) with $K/T = 0.5$, persistence $\rho = 0.9$, innovations rescaled to coordinate-wise standard deviation 0.2. $\nu = \infty$ denotes Gaussian innovations. Number of replications: 500.